

Mathematics Stage 4

Angle Relationships

Book 1 Applies angle relationships to solve problems, including those related to transversals on sets of parallel lines

Version: 260912

Feedback:
<https://MrDingMaths.com>

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Syllabus Content

MA4-ANG-C-01 applies angle relationships to solve problems, including those related to transversals on sets of parallel lines

Apply the language, notation and conventions of geometry

- Use appropriate terminology and conventions to define, label and name points, rays, lines and intervals using capital letters
- Identify and label the vertex and arms of an angle with capital letters
- Use appropriate conventions to label and name angles
- Use common conventions to indicate right angles, equal angles and intervals on diagrams

Identify geometrical properties of angles at a point

- Identify right angles, straight angles, angles of complete revolution and vertically opposite angles
- Apply the terms complementary and supplementary to a pair of angles adding to 90° and 180° , respectively
- Apply the term adjacent angles to a pair of angles with a common arm and common vertex

Identify and describe corresponding, alternate and co-interior angles when 2 straight lines are crossed by a transversal, including parallel lines

- Identify and describe perpendicular lines using the symbol for is perpendicular to ($\perp$)
- Apply the common conventions to indicate parallel lines on diagrams
- Identify and describe pairs of parallel lines using the symbol for is parallel to ($\parallel$)
- Identify and define transversals, including transversals of parallel lines
- Identify, name and measure alternate angle pairs, corresponding angle pairs and co-interior angle pairs for 2 lines cut by a transversal
- Verify and identify corresponding angles and alternate angles as equal, and co-interior angles as supplementary, when a pair of parallel lines is cut by a transversal
- Justify that 2 lines are parallel by using properties of alternate, corresponding or co-interior angles on parallel lines

Solve numerical problems involving angles using reasoning

- Apply the knowledge of angle relationships including angles at a point to find the sizes of unknown angles embedded in diagrams and give reasons
- Apply the knowledge of angles associated with parallel lines to find the sizes of unknown angles embedded in related diagrams and give reasons

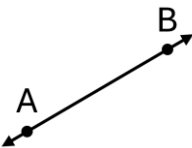
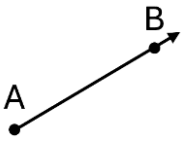
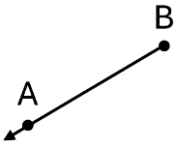
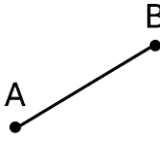
Points Lines Rays

i Building Blocks of Geometry

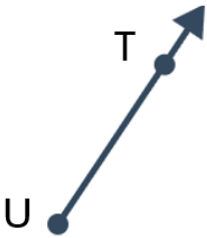
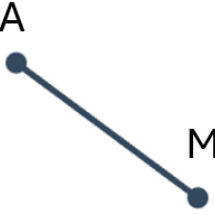
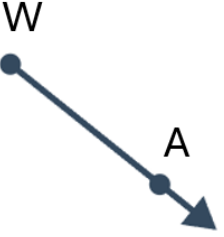
Geometry studies shapes, angles, and their properties.
Points, rays, lines, and intervals are geometry's building blocks.

Point	Line Segment	Ray	Line
			
Point A A point marks an exact location in space.	Segment AB or BA An interval or line segment joins two points.	Ray AB A ray starts at a point and goes on forever.	Line AB or BA A line passes through two points and extends forever.

? Identify line segments, rays, and lines

				
Line AB or BA	Ray AB	Ray BA	Interval AB or BA	Point B

Identify these geometrical objects.

a 	b 	c 	d 
e 	f 	g 	h 

1 Name these geometrical objects.

a

 T

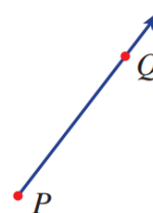
Point T

b



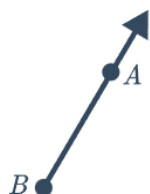
Line CD / DC

c



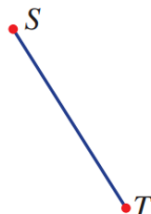
Ray PQ

d



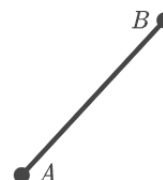
Ray BA

e



Interval ST / TS

f



Interval AB / BA

2 Draw these geometrical figures.

a

Line FG

b

Segment BA

c

Point C

d

Ray ZY

e

Ray YZ

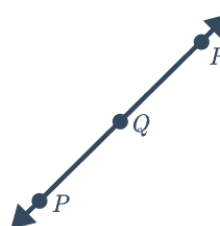
f

Line BA

3

a

Name a ray that is the same as the ray PQ .



Ray PR

b

Joseph says line segment PQ is the same as line segment PR . Explain why he is incorrect.

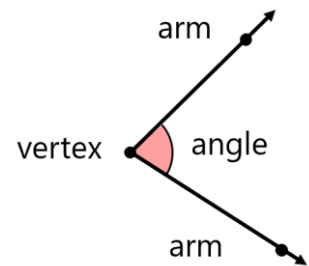
The line segments are different lengths.

Labelling Angles

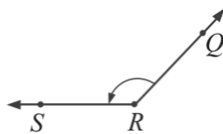
i Angles

Angles measure the turn between two intersecting lines.

- Lines **intersect** when they meet.
- The **vertex** is the point where the lines intersect.
- The intersecting lines are called **arms**.
- An **angle measures the turn** between arms.
- We show angles with a curved **arc**.

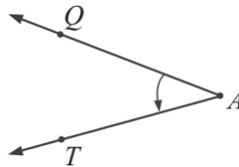


? Identify parts of an angle



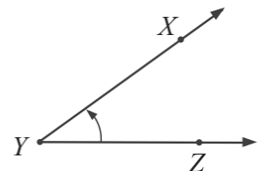
Vertex: **R**

Arms: **RS, RQ**



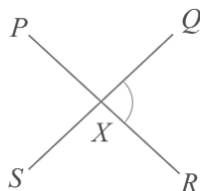
Vertex:

Arms:



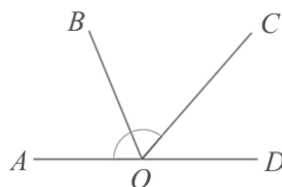
Vertex:

Arms:



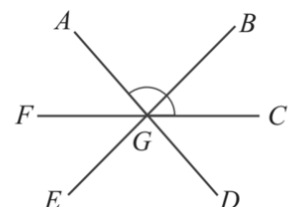
Vertex:

Arms:



Vertex:

Arms:



Vertex:

Arms:

💡 Key Ideas

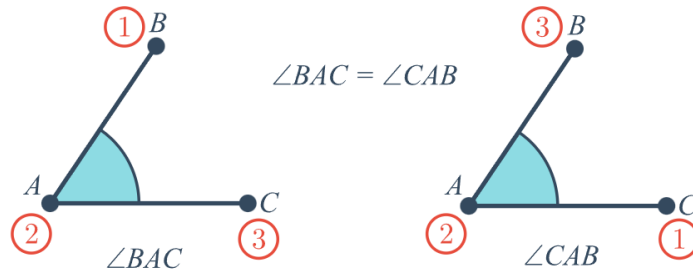
1. An angle forms when two lines
2. The point where two lines intersect is called the
3. The two lines that form the angle are called the

i Labelling Angles

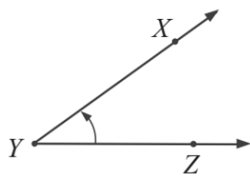
Label an angle using the angle symbol $\angle$ and three points.

Write the **vertex** as the **middle letter**.

The other two letters mark points on the **arms**.

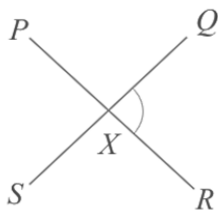


Example Label angles using three letters



$\angle XYZ$ or
 $\angle ZYX$

When naming an angle with three letters, the goes in the middle.



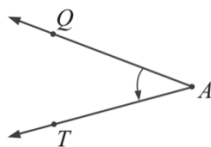
$\angle QXR$ or
 $\angle RXQ$

Why is labelling " $\angle X$ " not clear in this scenario?
There are angles at vertex X.

Guided Practice

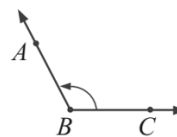
Label these angles.

a



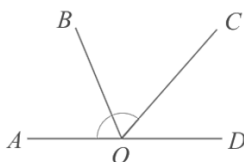
$\angle QAT$ or $\angle TAQ$

b



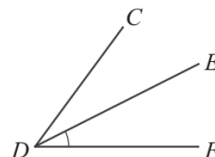
$\angle ABC$ or $\angle CBA$

c



$\angle AOC$ or $\angle COA$

d



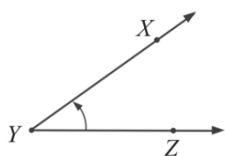
$\angle EDF$ or $\angle FDE$

Key Ideas

1. We name angles using letters.
2. We get these letters from the two of the angle and the

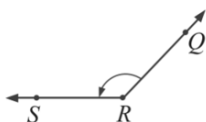
1 Label these angles.

a



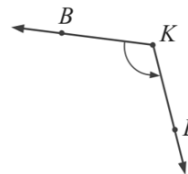
$\angle XYZ$ or $\angle ZYX$

b



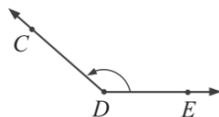
$\angle SRQ$ or $\angle QRS$

c



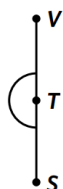
$\angle BKL$ or $\angle LKB$

d



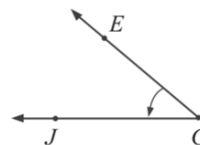
$\angle CDE$ or $\angle EDC$

e



$\angle VTS$ or $\angle STV$

f

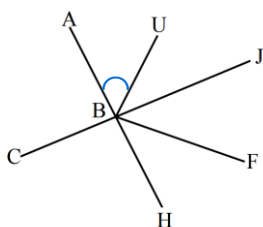


$\angle JGE$ or $\angle EGJ$

2 Draw an arc to show these angles on the diagram.

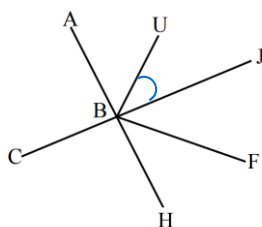
a

$\angle ABU$



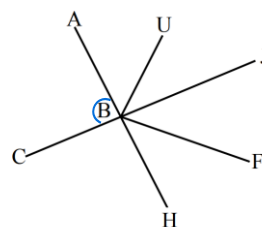
b

$\angle UBJ$



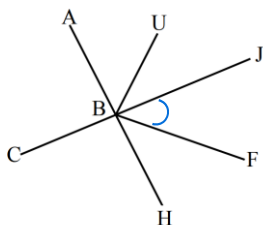
c

$\angle CBA$



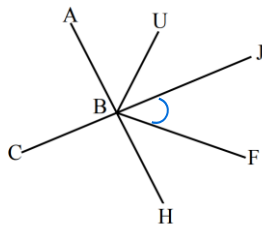
d

$\angle FBJ$



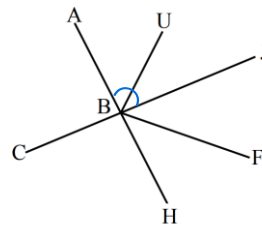
e

$\angle JBF$



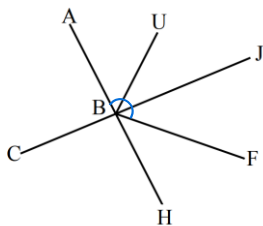
f

$\angle ABJ$



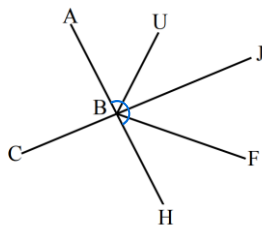
g

$\angle ABF$



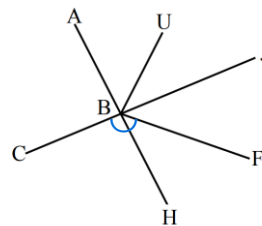
h

$\angle ABH$



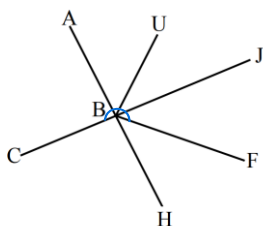
i

$\angle FBC$



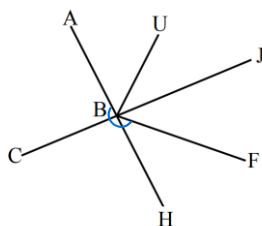
j

Reflex $\angle FBC$



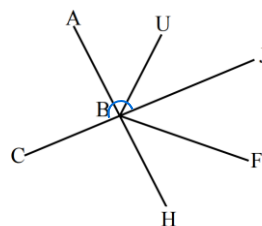
k

Reflex $\angle ABF$



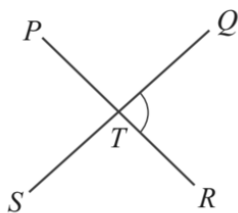
l

$\angle CBJ$



3 Complete the following to name the marked angle.

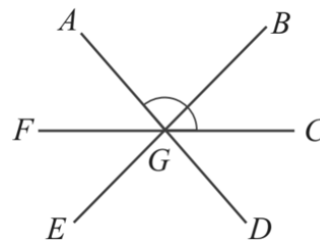
a



The vertex is
The arms are and
The angle is $\angle$

$\angle QTR$ or $\angle RTQ$

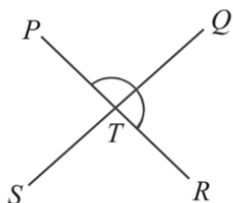
b



The vertex is
The arms are and
The angle is $\angle$

$\angle AGC$ or $\angle CGA$

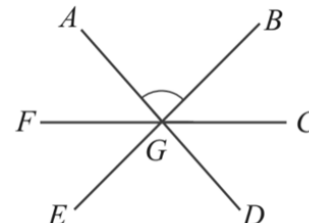
c



The vertex is
The arms are and
The angle is $\angle$

$\angle PTR$ or $\angle RTP$

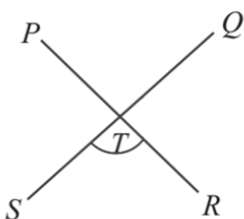
d



The vertex is
The arms are and
The angle is $\angle$

$\angle AGB$ or $\angle BGA$

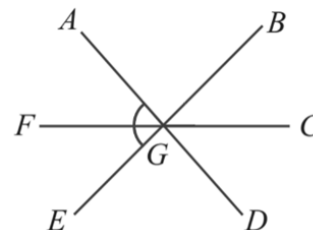
e



The vertex is
The arms are and
The angle is $\angle$

$\angle STR$ or $\angle RTS$

f



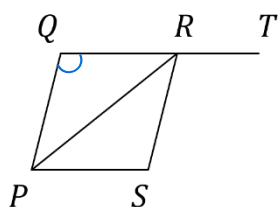
The vertex is
The arms are and
The angle is $\angle$

$\angle AGE$ or $\angle EGA$

4 Draw an arc to show these angles on the diagram.

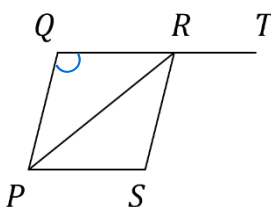
a

$\angle PQR$



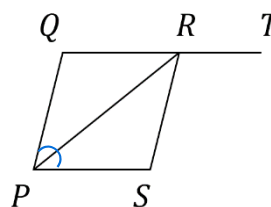
b

$\angle PQT$



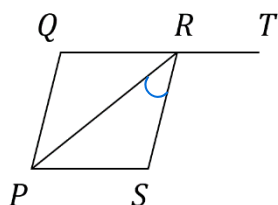
c

$\angle QPS$



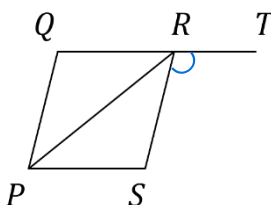
d

$\angle PRS$



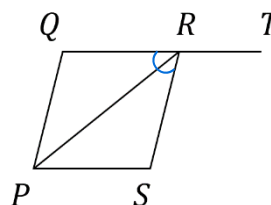
e

$\angle TRS$



f

$\angle SRQ$

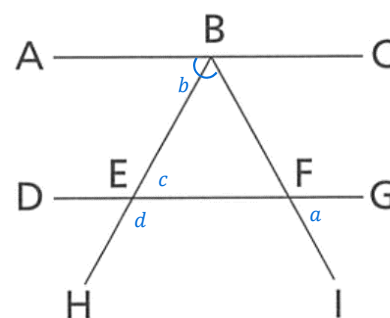


5 Label each of these angles on the diagram using a, b, c, d

a $\angle GFI$ b $\angle ABF$

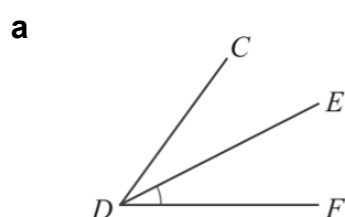
c $\angle BEF$ d $\angle FEH$

e Jackson says angles BEF and BEG are the same angle. Explain why he is correct.

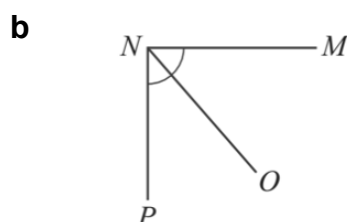


The angle is between the same arms EB and EF

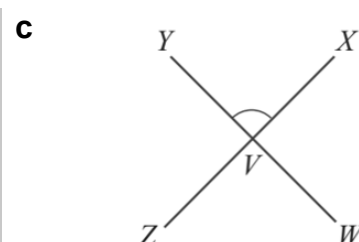
6 Name the angles shown by the arc.



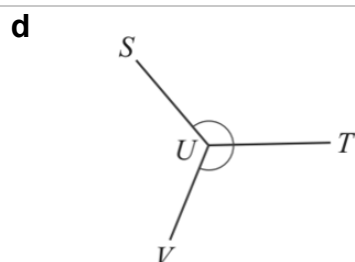
$\angle EDF$ or $\angle FDE$



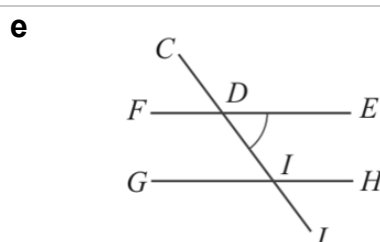
$\angle MNP$ or $\angle PNM$



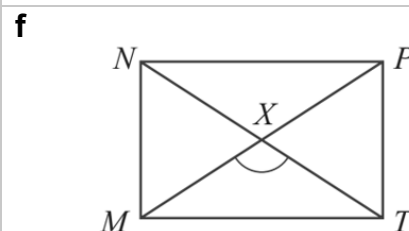
$\angle YVX$ or $\angle XVY$



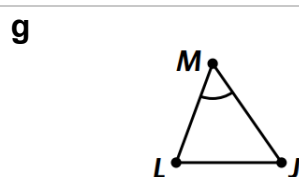
$\angle SUV$ or $\angle VUS$



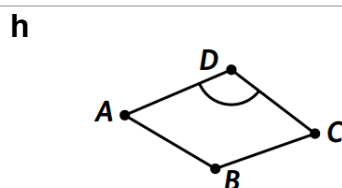
$\angle EDJ$ or $\angle JDE$



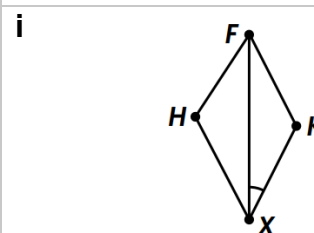
$\angle MXT$ or $\angle TXM$



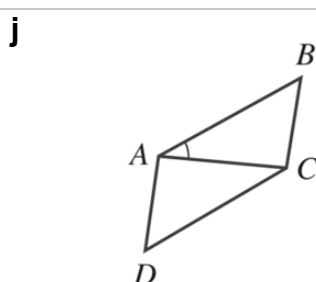
$\angle LMJ$ or $\angle JML$



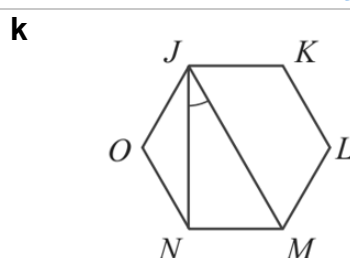
$\angle ADC$ or $\angle CDA$



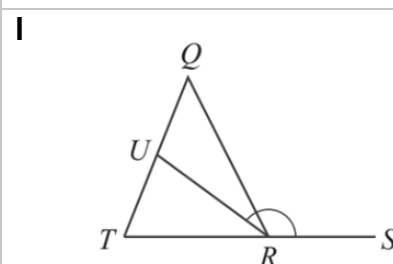
$\angle FXK$ or $\angle KXF$



$\angle BAC$ or $\angle CAB$



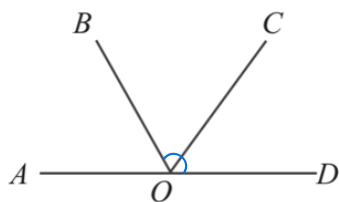
$\angle NJM$ or $\angle MJN$



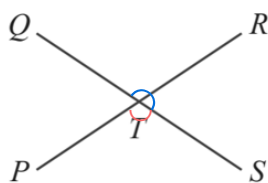
$\angle URS$ or $\angle SRU$

7 Mark these angles on the diagram.

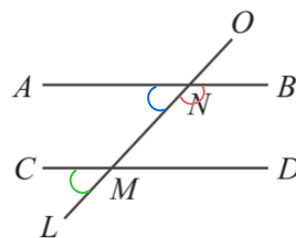
a $\angle BOD$



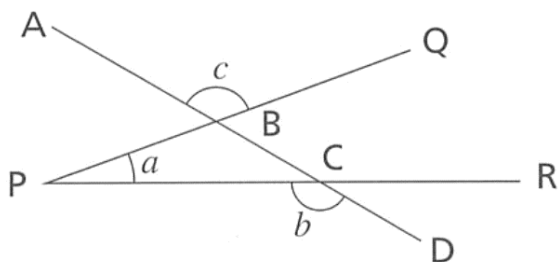
b $\angle QTS$ and $\angle STP$



c $\angle ANM, \angle MNB, \angle CML$



8 Give the three letter names of the angles labelled a, b and c.



a

$\angle QPR$

b

$\angle PCD$

c

$\angle ABQ$

9 In the diagram, find the sizes of

a $\angle AOB$

b $\angle BOC$

c $\angle AOC$

d $\angle AOD$

e $\angle BOD$

f $\angle DOC$

42°

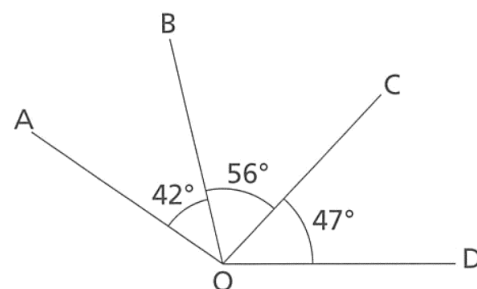
56°

98°

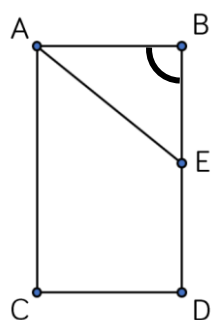
145°

103°

47°



10 A geometrical drawing is shown.



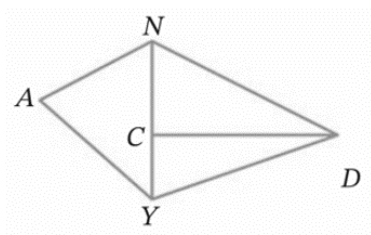
a Name three vertical line segments.

AC, BE, ED

b Write two different labels for the angle shown.

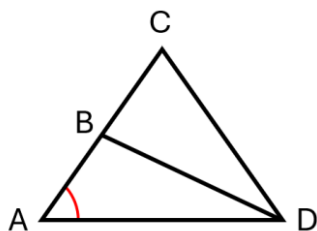
$\angle ABE$ or $\angle ABD$

11 Name three angles that have C as their vertex.



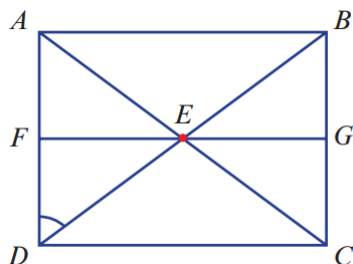
$\angle NCD, \angle DCY, \angle NCY$

- 12 Write four different names for the marked angle.



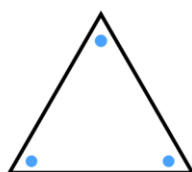
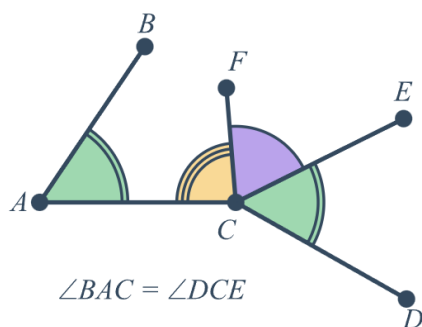
$\angle CAD, \angle BAD, \angle DAB, \angle DAC$

- 13 Write six different names for the marked angle.



$\angle ADB, \angle FDE, \angle ADE, \angle FDB, \angle BDA, \angle EDF, \angle EDA, \angle BDF$

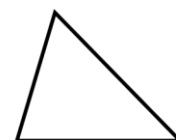
- 14 Equal angles are shown using the same number of arcs or using the same symbol.



3 angles equal

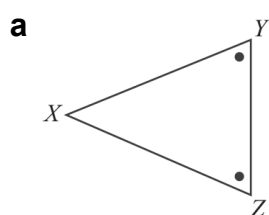


2 angles equal

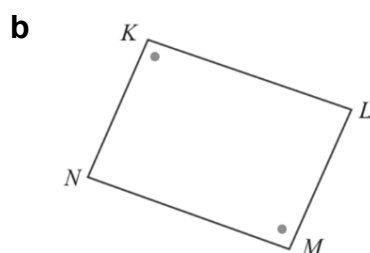


no angles equal

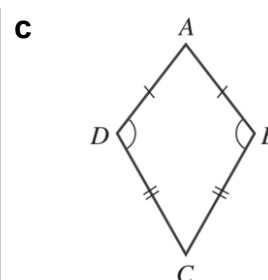
Name the pairs of equal angles.



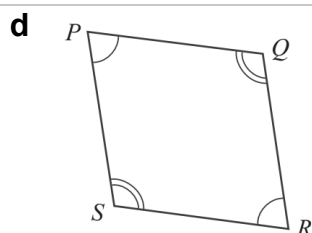
$\angle XYZ = \angle XZY$



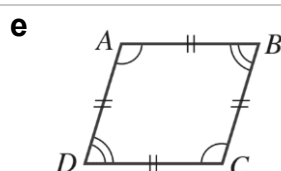
$\angle NKL = \angle LMN$



$\angle ADC = \angle ABC$



$\angle SPQ = \angle QRS$
 $\angle PSR = \angle PQR$



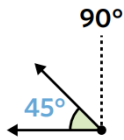
$\angle DAB = \angle BCD$
 $\angle ADC = \angle ABC$

Classifying Angles based on Turn

① Angles Classified Based on Turn

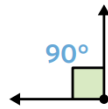
Acute angle

$$0^\circ < \theta < 90^\circ$$



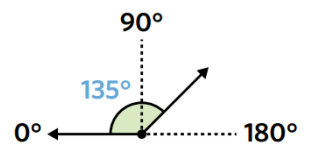
Right angle

$$\theta = 90^\circ$$



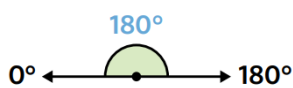
Obtuse angle

$$90^\circ < \theta < 180^\circ$$



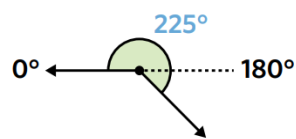
Straight line

$$\theta = 180^\circ$$



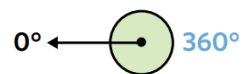
Reflex angle

$$180^\circ < \theta < 360^\circ$$



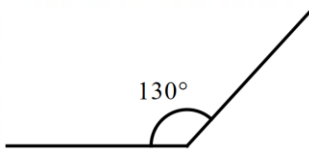
Full turn, or revolution

$$\theta = 360^\circ$$

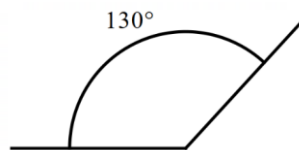


② Classify Angles

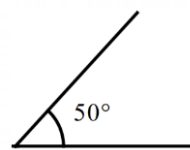
a



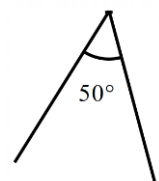
b



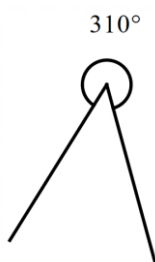
c



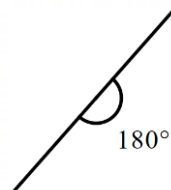
d



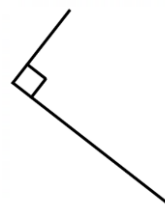
e



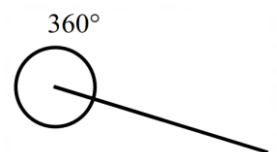
f



g

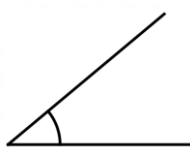


h



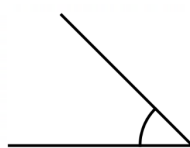
1 Decide whether each of these angles is acute, obtuse, or reflex.

a



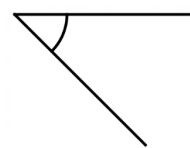
Acute Obtuse Reflex
acute

b



Acute Obtuse Reflex
acute

c



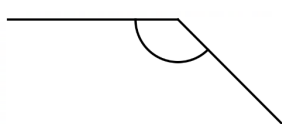
Acute Obtuse Reflex
acute

d



Acute Obtuse Reflex
obtuse

e



Acute Obtuse Reflex
obtuse

f



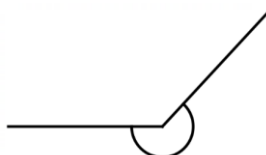
Acute Obtuse Reflex
obtuse

g



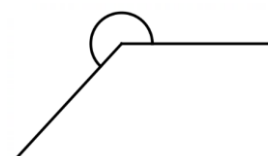
Acute Obtuse Reflex
obtuse

h



Acute Obtuse Reflex
reflex

i



Acute Obtuse Reflex
reflex

j



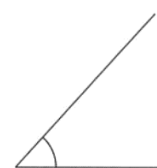
Acute Obtuse Reflex
acute

k



Acute Obtuse Reflex
obtuse

l



Acute Obtuse Reflex
acute

m



Acute Obtuse Reflex
acute

n



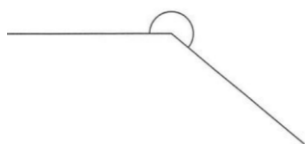
Acute Obtuse Reflex
obtuse

o



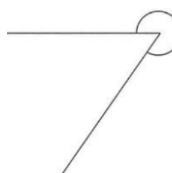
Acute Obtuse Reflex
obtuse

p



Acute Obtuse Reflex
reflex

q



Acute Obtuse Reflex
reflex

r

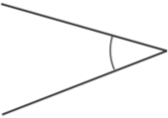
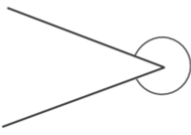
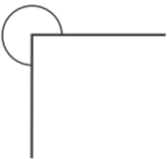
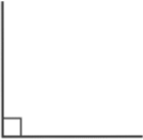
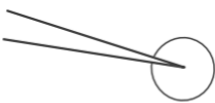
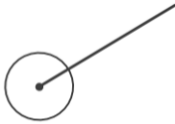


Acute Obtuse Reflex
reflex

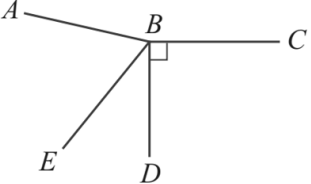
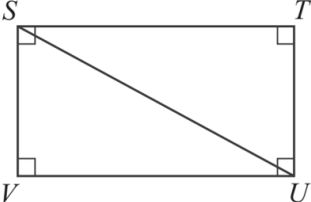
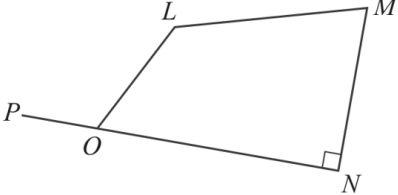
2 Classify these angles.

a 58° acute	b 175° obtuse	c 90° right angle	d 195° reflex
e 360° revolution	f 149° obtuse	g 242° reflex	h 180° straight line
i 15° acute	j 136° obtuse	k 305° reflex	l 224° reflex

3 Classify these angles.

a  acute	b  obtuse	c  straight line
d  right angle	e  reflex	f  obtuse
g  reflex	h  right angle	i  reflex
j  reflex	k  straight line	l  revolution

4 Classify the angles shown in these diagrams as either acute, obtuse, or right angle.

a 	b 	c 
$\angle ABC$ obtuse	$\angle SVU$ right angle	$\angle OLM$ obtuse
$\angle CBD$ right angle	$\angle STU$ right angle	$\angle PON$ straight line
$\angle EBD$ acute	$\angle VSU$ acute	$\angle LOP$ obtuse
$\angle ABE$ acute	$\angle TUS$ acute	$\angle ONM$ right angle

7 Look at the letters below. Which one has both parallel and perpendicular lines?

A C E L Z

E

8 Which of these angles:

105°, 95°, 25°, 85°, 175°

a Is the smallest obtuse angle?

95°

b Is the largest acute angle?

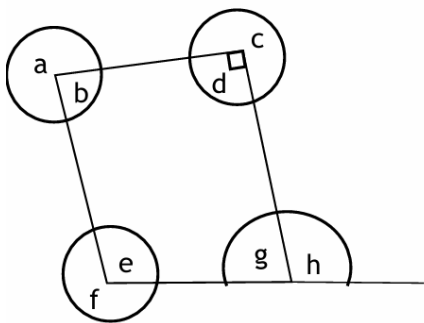
85°

9 Which of these angles are:

a Acute

b Obtuse

c Reflex



b, g

e, h

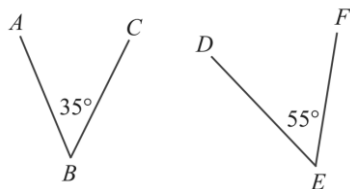
a, c, f

Complementary and Supplementary Angles

① Complementary and Supplementary Angles

Complementary angles

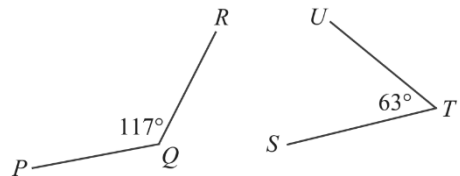
If two angles **sum to 90°** , they are complementary



$$\angle ABC + \angle DEF = 90^\circ$$

Supplementary angles

If two angles **sum to 180°** , they are supplementary



$$\angle PQR + \angle SUT = 180^\circ$$

② Identify whether two angles are complementary, supplementary, or neither

a

$140 + 40 = \dots\dots\dots$

- ☐ Complementary
☐ Supplementary
☐ Neither

b

$\dots\dots\dots + \dots\dots\dots = \dots\dots\dots$

- ☐ Complementary
☐ Supplementary
☐ Neither

c

$\dots\dots\dots + \dots\dots\dots = \dots\dots\dots$

- ☐ Complementary
☐ Supplementary
☐ Neither

d

$\dots\dots\dots + \dots\dots\dots = \dots\dots\dots$

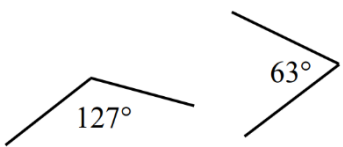
- ☐ Complementary
☐ Supplementary
☐ Neither

💡 Key Ideas

- Two angles that sum to $\dots\dots\dots$ are supplementary.
- Two angles that sum to $\dots\dots\dots$ are complementary.

1 Decide whether these angles are complementary, supplementary, or neither.

a

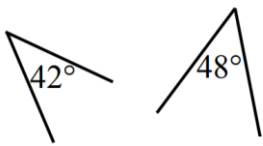


..... + =

☐ Complementary
☐ Supplementary
☐ Neither

neither

b

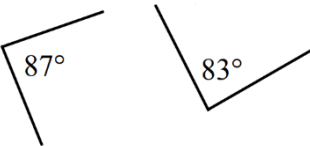


..... + =

☐ Complementary
☐ Supplementary
☐ Neither

complementary

c

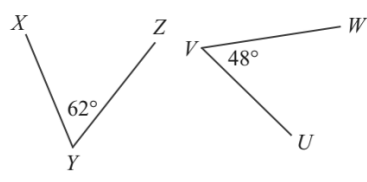


..... + =

☐ Complementary
☐ Supplementary
☐ Neither

neither

d

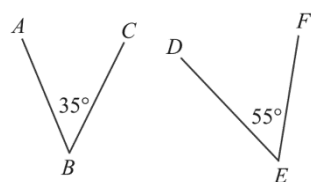


..... + =

☐ Complementary
☐ Supplementary
☐ Neither

neither

e

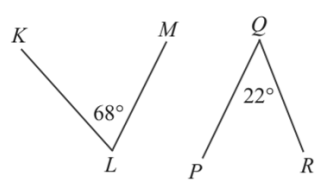


..... + =

☐ Complementary
☐ Supplementary
☐ Neither

complementary

f

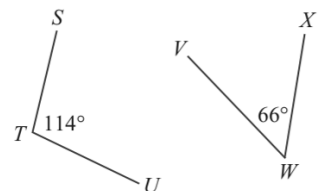


..... + =

☐ Complementary
☐ Supplementary
☐ Neither

complementary

g

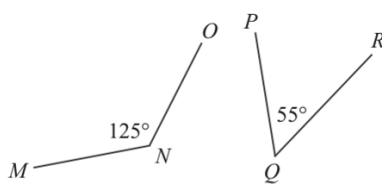


..... + =

☐ Complementary
☐ Supplementary
☐ Neither

supplementary

h

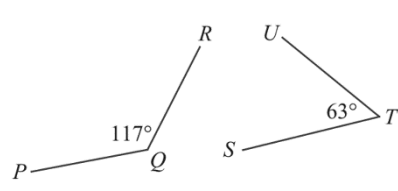


..... + =

☐ Complementary
☐ Supplementary
☐ Neither

supplementary

i



..... + =

☐ Complementary
☐ Supplementary
☐ Neither

supplementary

2 When two angles add up to 90° , we call each angle the “complement” of the other. Find the complement of these angles by subtracting it from 90° .

a 30°

60°

b 45°

45°

c 73°

17°

d 21°

69°

e 9°

81°

f 57°

33°

3 When two angles add up to 180° , we call each angle the “supplement” of the other. Find the supplement of these angles by subtracting it from 180° .

a 110°

70°

b 45°

135°

c 162°

18°

d 16°

164°

e 123°

57°

f 27°

153°

4 Complete the table.

Angle	Complement	Supplement
25°	65°	155°
65°	25°	115°
13°	77°	167°
79°	11°	101°
57°	33°	123°
3°	87°	177°
41°	49°	139°

5 Work out the:

a complement of 40° 50°	b supplement of 60° 120°	c complement of 41° 49°
d supplement of 100° 80°	e supplement of 15° 165°	f supplement of 48° 132°
g complement of 72° 18°	h complement of 5° 85°	i supplement of 90° 90°

6 If a pair of complementary angles are placed adjacent to each other, what type of angle is formed?

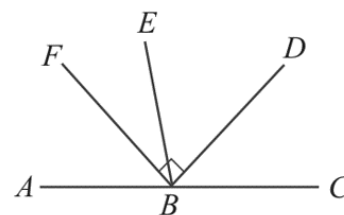
Right-angle

7 If a pair of supplementary angles are placed adjacent to each other, what type of angle is formed?

Straight line

8 State whether these pairs of angles are complementary, supplementary, or neither.

a $\angle FBE$ and $\angle DBE$ complementary	b $\angle ABE$ and $\angle CBE$ supplementary
c $\angle FBE$ and $\angle DBC$ neither	d $\angle FBC$ and $\angle FBA$ supplementary
e $\angle FBA$ and $\angle DBC$ neither	f $\angle FBA$ and $\angle EBD$ neither



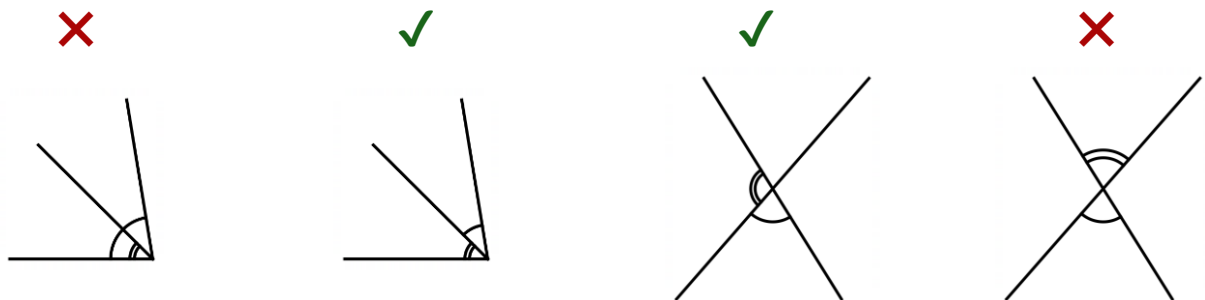
Adjacent Angles

① Adjacent Angles

Adjacent means “next to”.

If two angles share a **common vertex**, a **common arm**, and **do not overlap**, then they are adjacent angles.

② Identify whether two angles are adjacent



Decide whether these marked angles are adjacent.

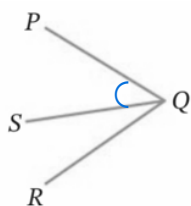
a	b	c
d	e	f

💡 Key Ideas

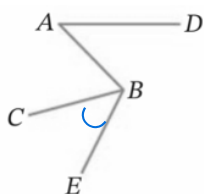
- Angles that are ‘next to’ each other are called, they have a common and

1 On the diagram, identify an angle adjacent to:

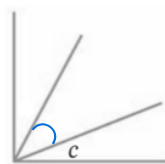
a $\angle RQS$



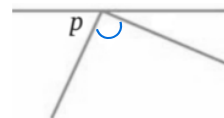
b $\angle ABC$



c Angle marked c

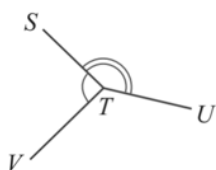


d Angle marked p



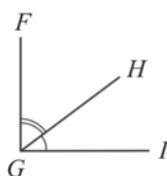
2 Decide whether the marked angles in each diagram are adjacent.

a



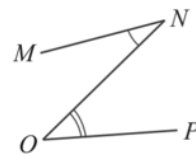
adjacent

b



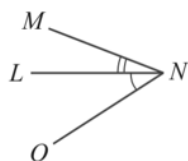
adjacent

c



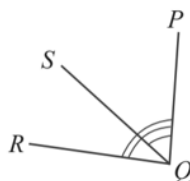
not adjacent

d



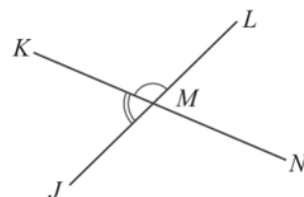
adjacent

e



Not adjacent

f



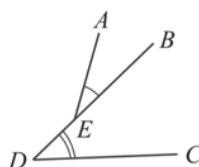
adjacent

g



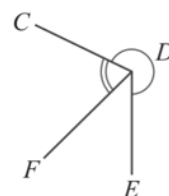
adjacent

h



Not adjacent

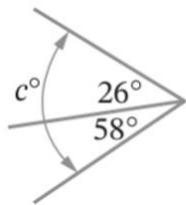
i



adjacent

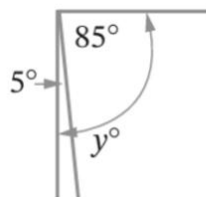
3 Find the value of the pronumeral in each diagram.

a



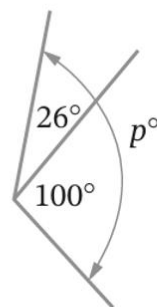
84°

b



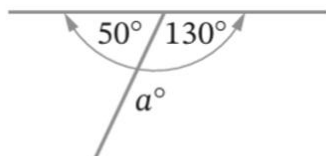
90°

c



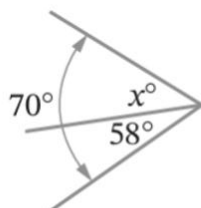
126°

d



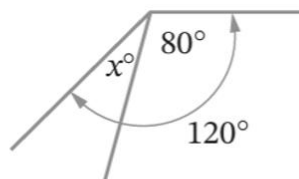
180°

e



12°

f



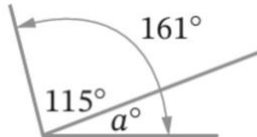
40°

g



15°

h



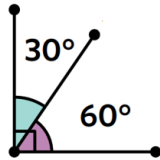
46°

Angles in a Right Angle and Straight Line

i Angles in a Right-Angle and Straight Line

Angles in a right-angle

If angles are within a right angle, then they add up to 90 degrees.



<https://www.geogebra.org/m/htq7jgyq>

To find unknown angles involving angles in a right-angle and straight line:

1. Write an equation using angle relationships.
2. Simplify the equation, if needed.
3. Solve for the unknown.

Angles on a straight line

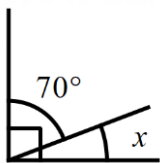
If angles are on a straight-line, then they add up to 180 degrees.



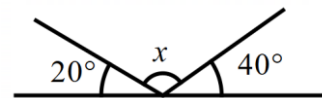
<https://www.geogebra.org/m/bfdfrbqg>

Example Solve problems involving angles in a right-angle and straight line

Find the value of the pronumerals.



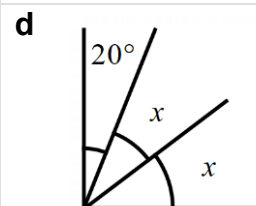
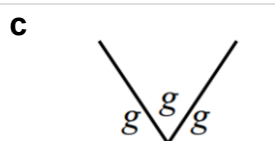
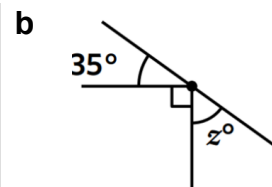
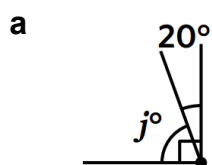
1. $x + 70 = 90$ (angles in a right-angle)
3. $x = 20$



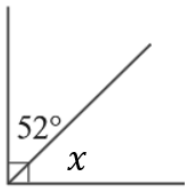
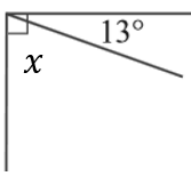
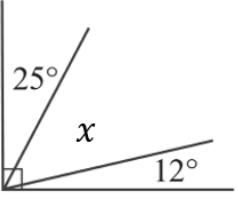
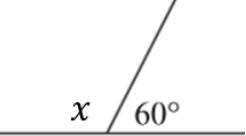
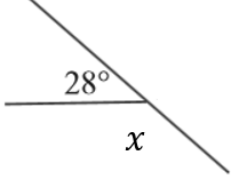
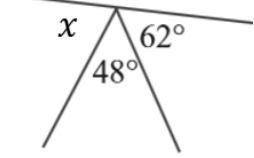
1. $x + 20 + 40 = 180$ (angles on a straight-line)
2. $x + 60 = 180$
3. $x = 120$

Guided Practice

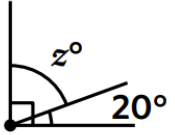
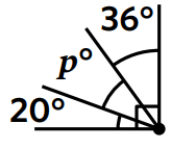
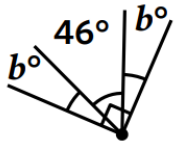
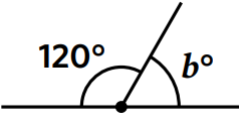
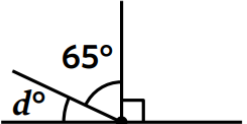
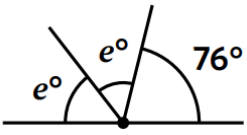
Calculate the value of the pronumerals.



1 Find the value of each pronumeral.

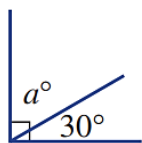
a  $x + 52 = 90$ $x = \dots\dots$ 38°	b  $x + \dots\dots = \dots\dots$ $x = \dots\dots$ 77°	c  $x + \dots\dots + \dots\dots = \dots\dots$ $x + \dots\dots = \dots\dots$ $x = \dots\dots$ 53°
d  $x + 60 = 180$ $x = \dots\dots$ 120°	e  $x + \dots\dots = \dots\dots$ $x = \dots\dots$ 152°	f  $x + \dots\dots + \dots\dots = \dots\dots$ $x + \dots\dots = \dots\dots$ $x = \dots\dots$ 70°

2 Calculate the value of the unknown angle.

a  $z + 20 = 90$ $z = \dots\dots\dots$ 70°	b  $p + 36 + 20 = 90$ $p = \dots\dots\dots$ 34°	c  $b + 46 + b = 90$ $b = \dots\dots\dots$ 22°
d  $b + 120 = 180$ $b = \dots\dots\dots$ 60°	e  $d + 65 = 90$ $d = \dots\dots\dots$ 25°	f  $e + e + 76 = 180$ $e = \dots\dots\dots$ 52°

3 Calculate the value of the letter a in each diagram, giving reasons.

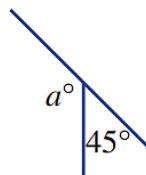
a $a + 30 = 90$ (angles in a right angle)



$a = \dots\dots$

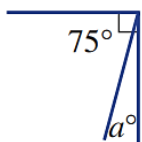
60°

b



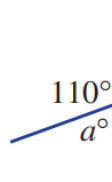
$a + 45 = 180$ (angles on a straight line)
 $a = 135$

c



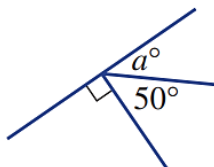
$a + 75 = 90$ (angles in a right angle)
 $a = 15$

d



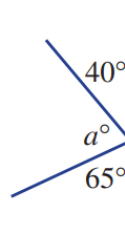
$a + 110 = 180$ (angles in a straight line)
 $a = 70$

e



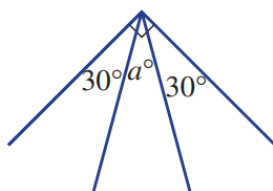
$a + 50 + 90 = 180$ (angles in a straight line)
 $a + 140 = 180$
 $a = 40$

f



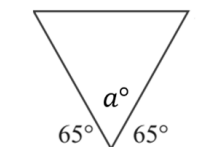
$a + 40 + 65 = 180$ (angles in a straight line)
 $a + 105 = 180$
 $a = 75$

g



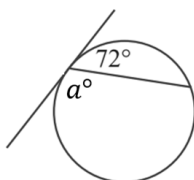
$a + 30 + 30 = 90$ (angles in a right angle)
 $a + 60 = 90$
 $a = 30$

h



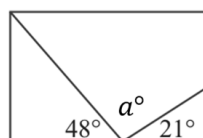
$a + 65 + 65 = 180$ (angles in a straight line)
 $a + 130 = 180$
 $a = 50$

i



$a + 72 = 180$ (angles in a straight line)
 $a = 108$

j



$a + 48 + 21 = 180$ (angles in a straight line)
 $a + 69 = 180$
 $a = 111$

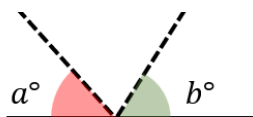
- 4 Marta thinks that $x = 180^\circ - 41^\circ = 139^\circ$ because angles on a straight line add up to 180° . Explain why she is incorrect.



41° and x do not share a common vertex; they are not adjacent.

- 5 Decide whether each statement is true or false.

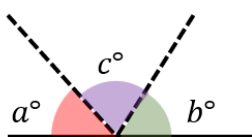
a $a + b = 180$



True False

false

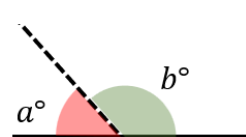
b $a + b + c = 180$



True False

true

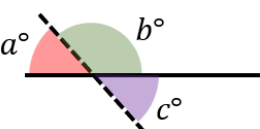
c $a + b = 180$



True False

true

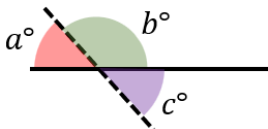
d $a + b + c = 180$



True False

false

e $b + c = 180$



True False

true

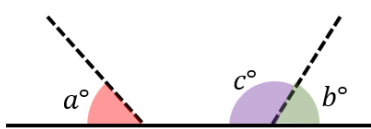
f $a + b = 180$



True False

false

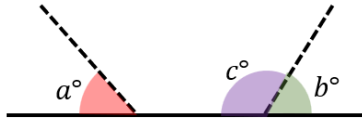
g $a + b + c = 180$



True False

false

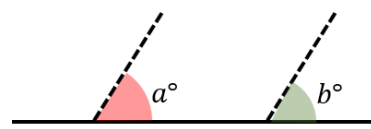
h $b + c = 180$



True False

true

i $a + b = 180$



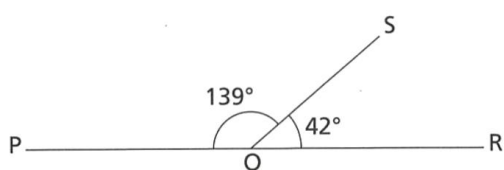
True False

false

- 6 Felix has a gaming chair that has a backrest that can lean all the way back until it is flat. If the angle between the chair and the backrest is 119° , how much more will it have to turn to lay flat?

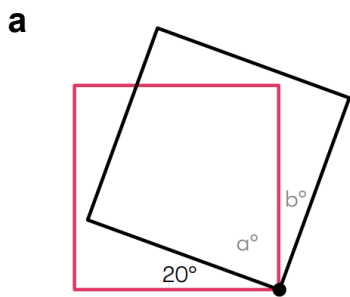
61°

- 7 Explain why PQR is not a straight line.

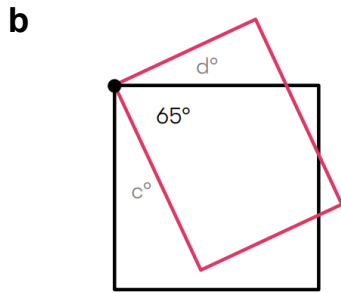


$139 + 42 = 181^\circ$, a straight line must sum to 180

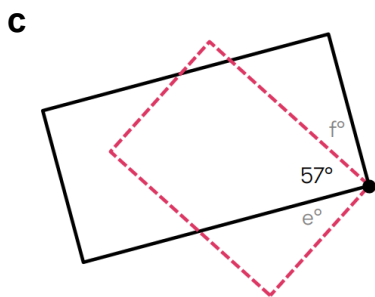
8 Determine the size of the missing angles using complementary angles.



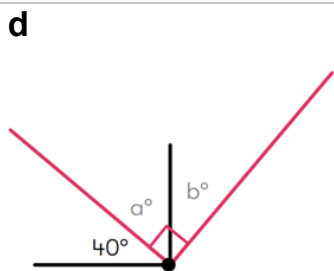
$a = 70, b = 20$



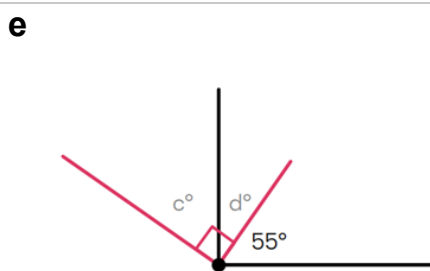
$c = 25, d = 25$



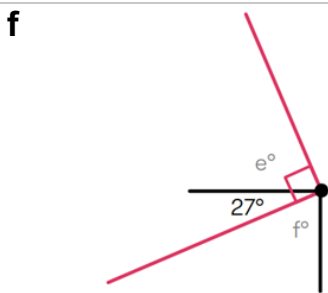
$e = 33, f = 33$



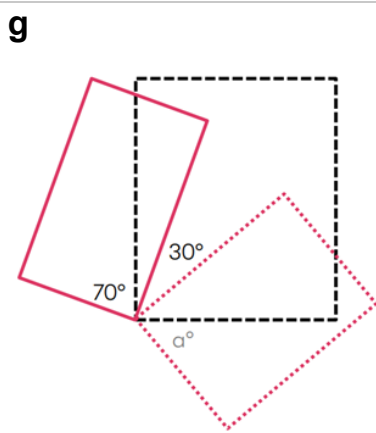
$a = 50, b = 40$



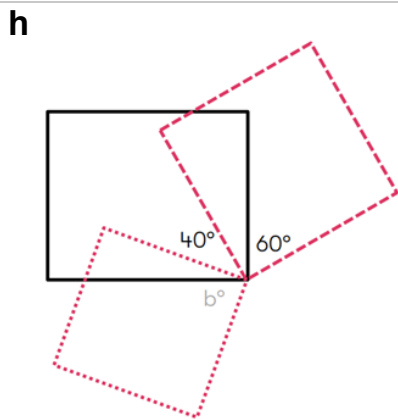
$c = 55, d = 35$



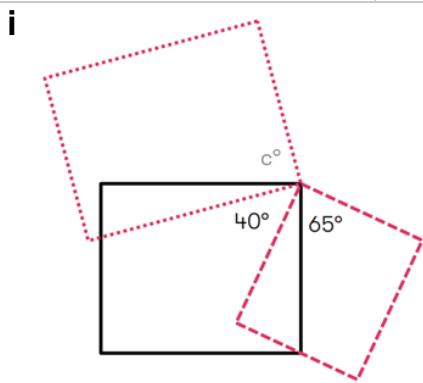
$e = 63, f = 63$



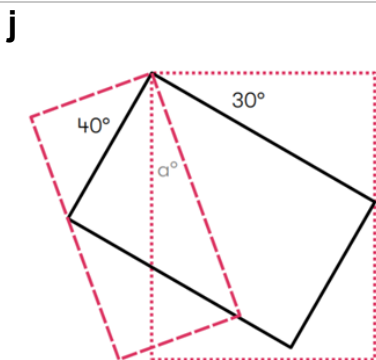
$a = 50$



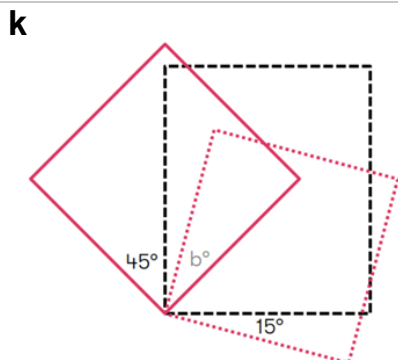
$b = 70$



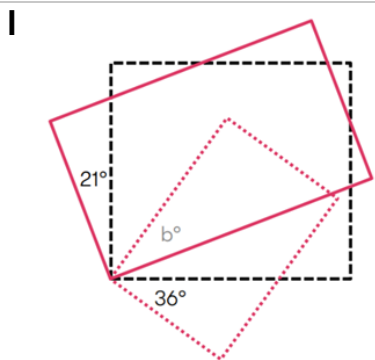
$c = 65$



$a = 20$



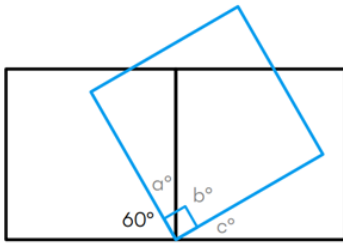
$a = 30$



$b = 33$

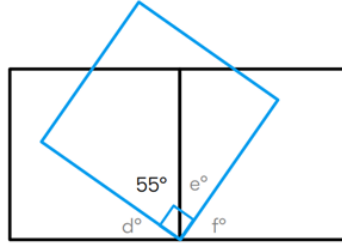
9 Determine the size of the missing angles using supplementary angles.

a



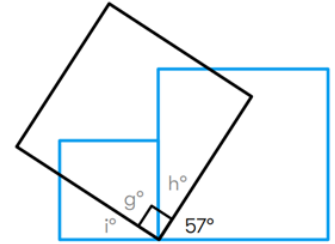
$$a = 30, b = 60, c = 30$$

b



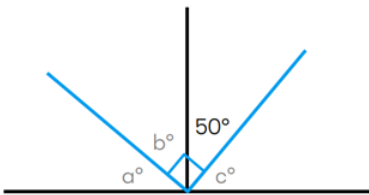
$$d = 35, e = 35, f = 55$$

c



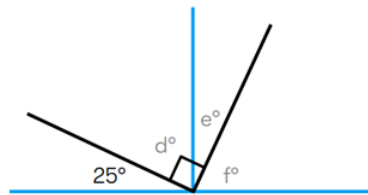
$$g = 57, h = 33, i = 33$$

d



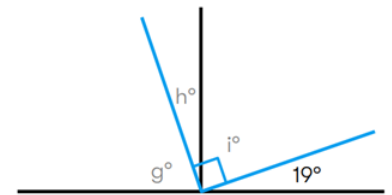
$$a = 50, b = 40, c = 40$$

e



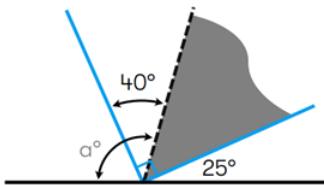
$$d = 65, e = 25, f = 65$$

f



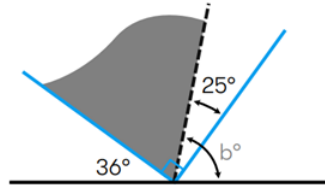
$$g = 71, h = 19, i = 71$$

g



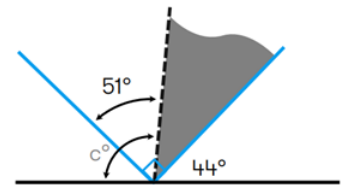
$$a = 105$$

h



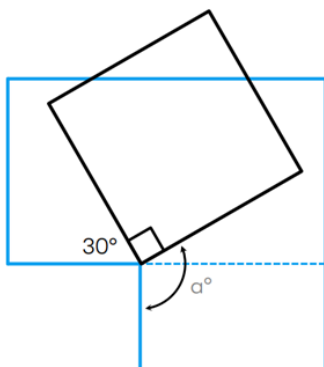
$$b = 79$$

i



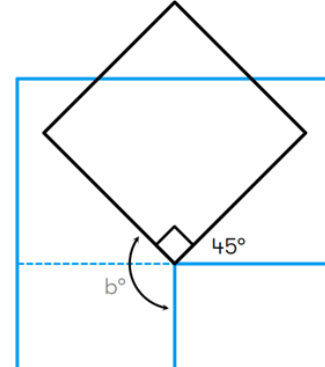
$$c = 97$$

j



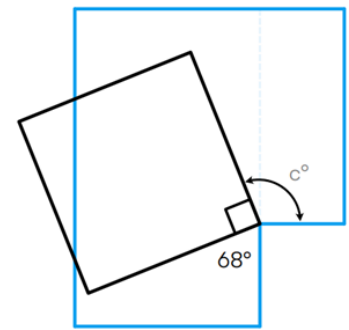
$$a = 150$$

k



$$b = 135$$

l

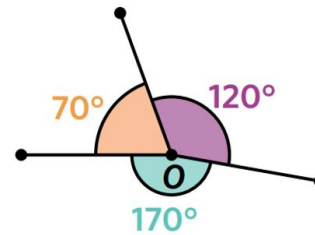


$$c = 112$$

Angles at a Point

i Angles at a Point

Angles that share a common vertex sum to 360°

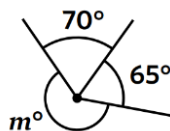


To find unknown angles involving angles at a point:

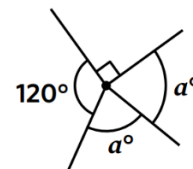
1. Write an equation using angle relationships.
2. Simplify the equation, if needed.
3. Solve for the unknown.

Example Solve problems involving angles at a point

Find the value of the pronumerals.



1. $m + 70 + 65 = 360$ (angles at a point)
2. $m + 135 = 360$
3. $m = 225$

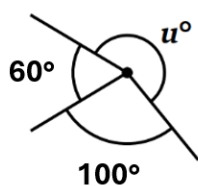


1. $a + a + 120 + 90 = 360$ (angles at a point)
2. $2a + 210 = 360$
3. $2a = 150$
 $a = 75$

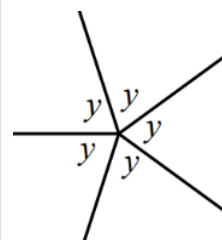
Guided Practice

Find the value of the pronumerals.

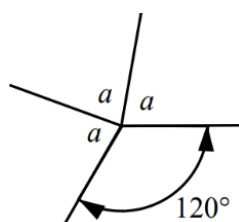
a



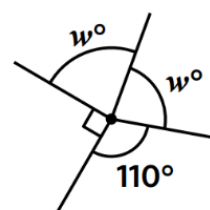
b



c

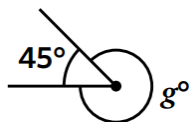


d



1 Calculate the value of the unknown angle.

a



$$45 + g = 360$$

$$g = \dots\dots\dots$$

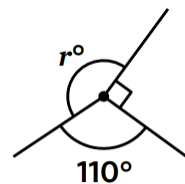
315°

b



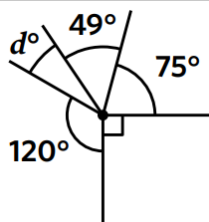
330°

c



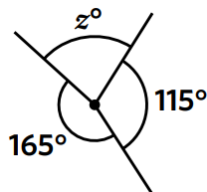
160°

d



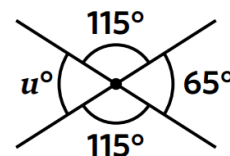
26°

e



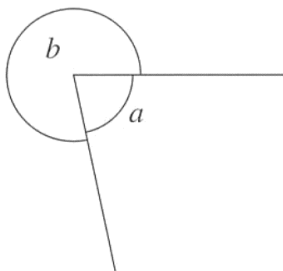
80°

f



65°

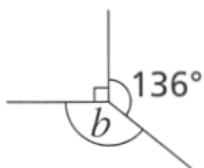
2 Seb measures these angles and writes $a = 82^\circ$ and $b = 284^\circ$. Explain how you know that Seb must have measured incorrectly.



$$a + b \text{ must sum to } 360. 284 + 82 = 366.$$

3 Calculate the value of the pronumeral.

a



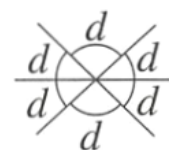
134°

b



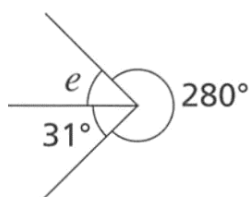
90°

c



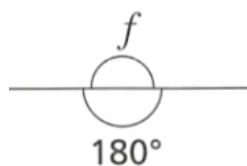
60°

d



49°

e



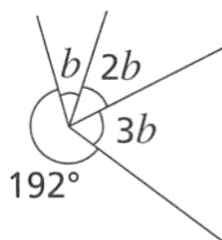
180°

f



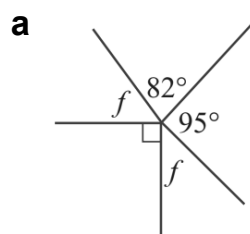
138°

4 Determine the value of b .

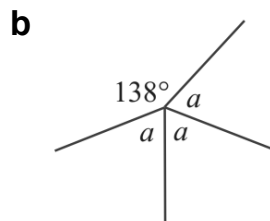


28°

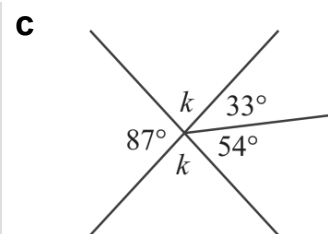
5 Calculate the value of the pronumeral.



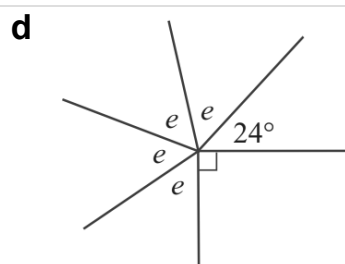
46.5°



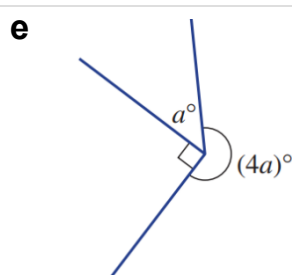
74°



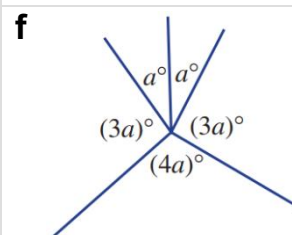
93°



61.5°

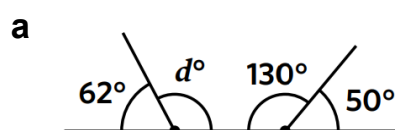


54°

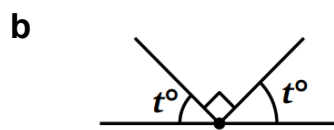


30°

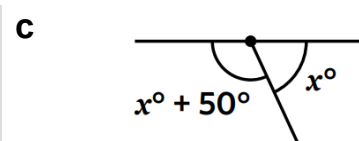
10 Calculate the unknown angles.



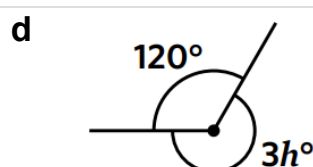
118°



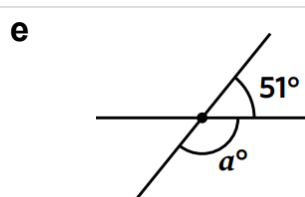
45°



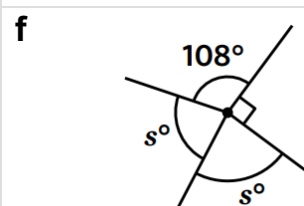
65°



80°



129°

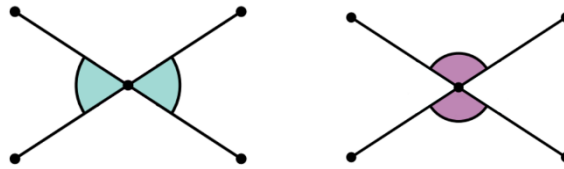


81°

Vertically Opposite Angles

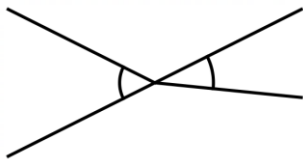
i Vertically Opposite Angles

Vertically opposite angles are opposite each other at the vertex when two lines intersect.

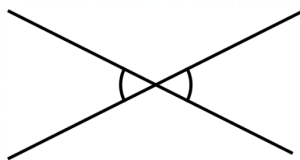


? Identify the vertically opposite angle

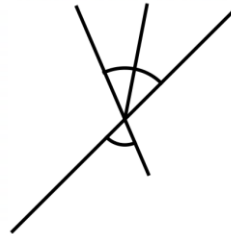
✗



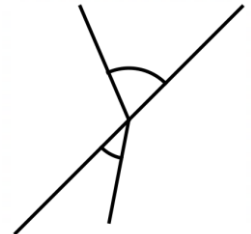
✓



✓

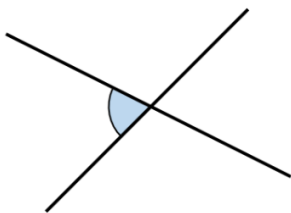


✗

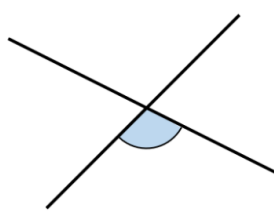


Identify the angle that is vertically opposite angle to the identified angle.

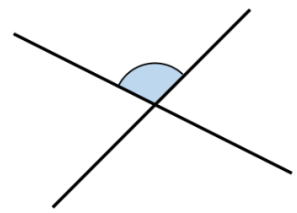
a



b



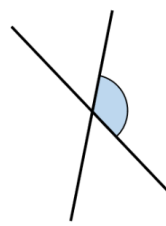
c



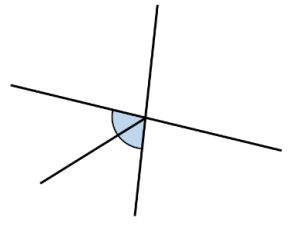
d



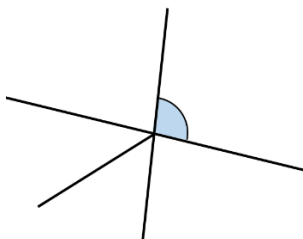
e



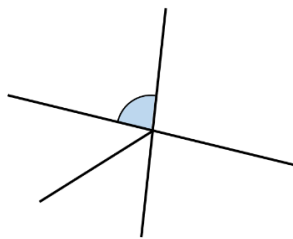
f



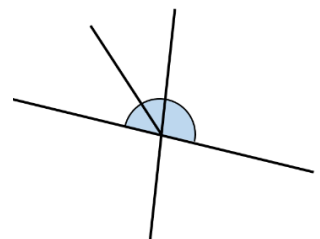
g



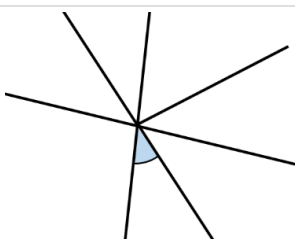
h



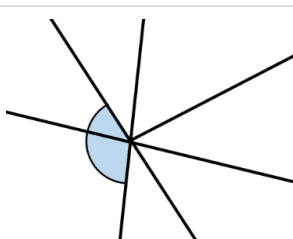
i



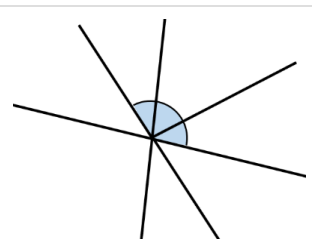
j



k

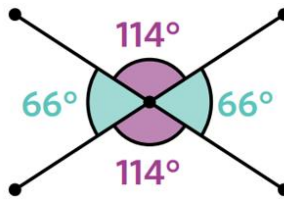


l



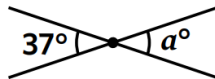
i Vertically Opposite Angles are Equal

Vertically opposite angles are equal.



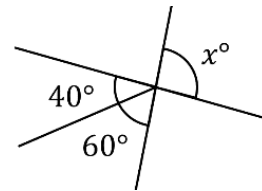
Example Solve problems involving vertically opposite angles

Calculate the value of a



$$a = 37 \text{ (vertically opposite)}$$

Calculate the value of x

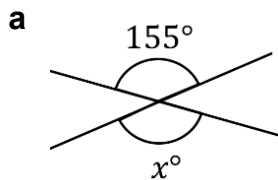


$$40 + 60 = x \text{ (vertically opposite)}$$

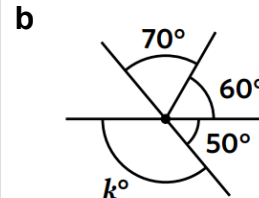
$$100 = x$$

Guided Practice

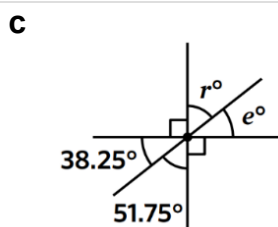
Calculate the value of the pronumerals.



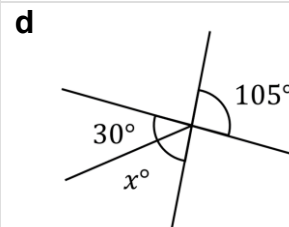
$$x = 155^\circ$$



$$k = 130^\circ$$



$$r = 51.75^\circ, e = 38.25^\circ$$

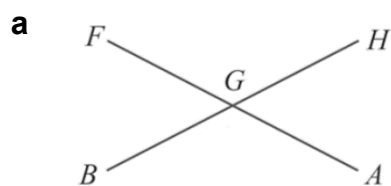


$$x = 75^\circ$$

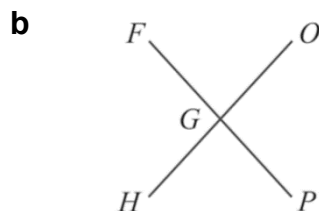
Key Ideas

1. Vertically opposite angles are

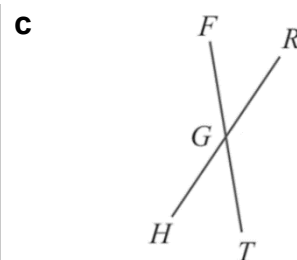
1 Identify the vertically opposite angle to $\angle FGH$ in these diagrams.



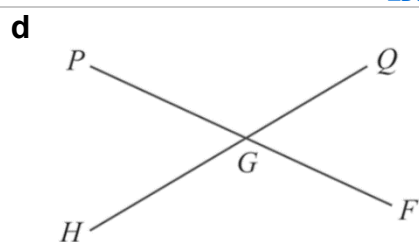
$\angle BGA$



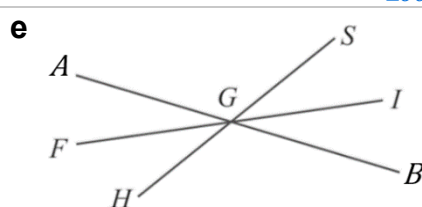
$\angle OGP$



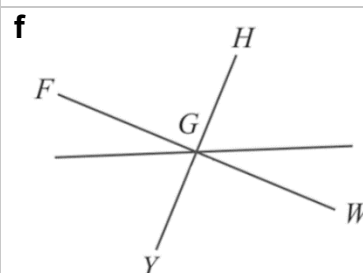
$\angle RGT$



$\angle PGQ$

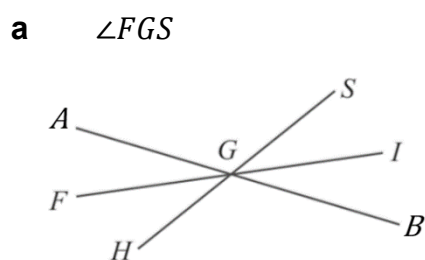


$\angle SGI$

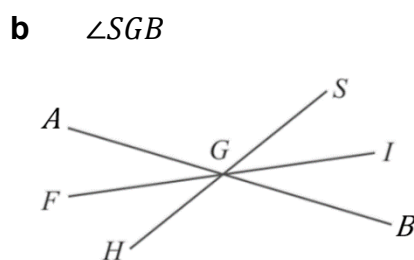


$\angle YGW$

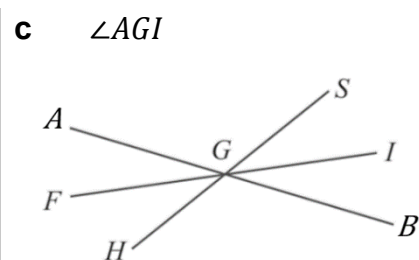
2 Label the angle and the angle vertically opposite to it.



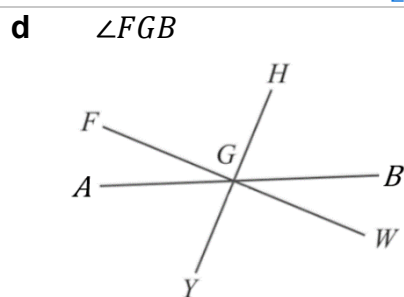
$\angle HGI$



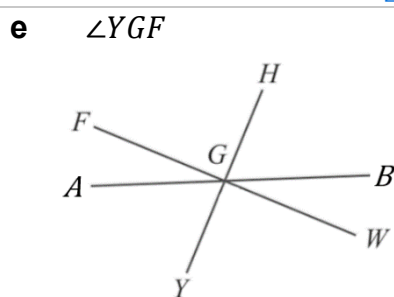
$\angle AGH$



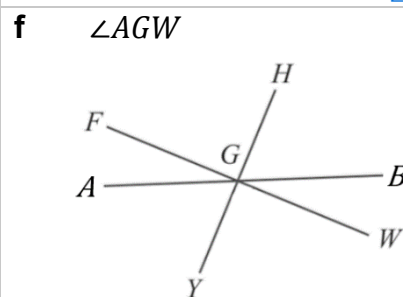
$\angle FGB$



$\angle AGW$



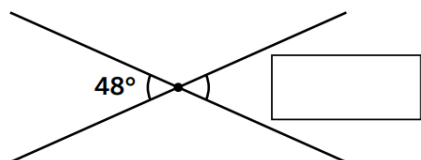
$\angle BGA$



$\angle FGB$

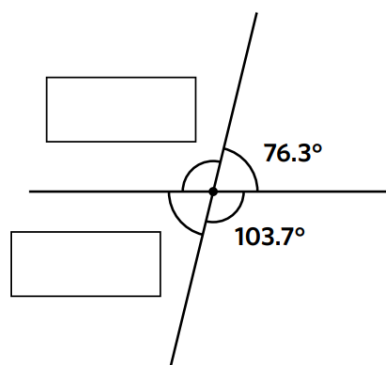
3 Calculate the value of the unknown angles.

a



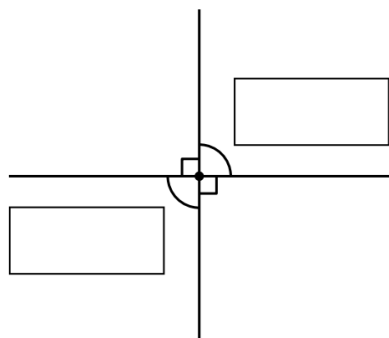
48°

b



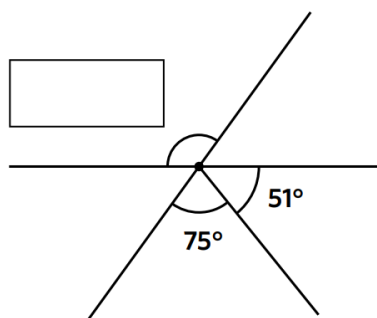
103.7°, 76.3°

c



90°, 90°

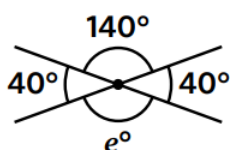
d



126°

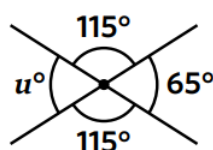
4 Determine the value of each unknown angle.

a



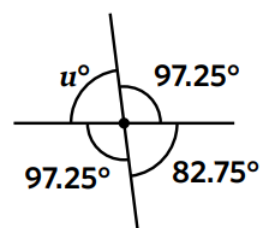
140°

b



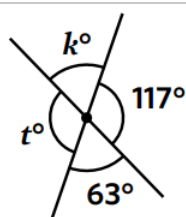
65°

c



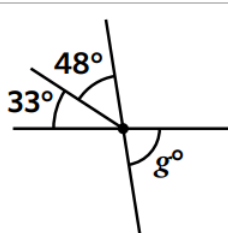
82.75°

d



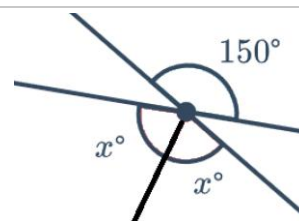
$t = 117^\circ$
 $k = 63^\circ$

e



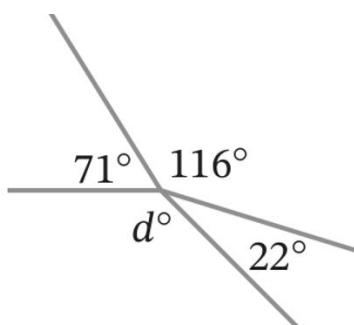
81°

f



75°

5 Jimmy says $d = 116^\circ$ because it is vertically opposite to 116° . Explain why he is incorrect and find the correct value of d .



d is not vertically opposite because it is not formed by the same straight lines that form 116°

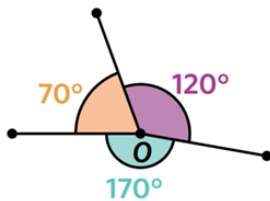
We can find d using angles at a point; $360 - 22 - 116 - 71 = 151$

Mixed Practice

i Angle Relationships

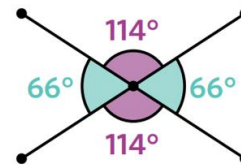
Angles at a Point

If angles share a common vertex, then they sum to 360°



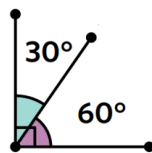
Vertically Opposite Angles

If angles are vertically opposite, then they are equal



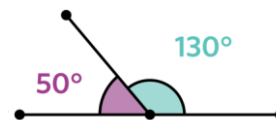
Angles in a Right-Angle

If angles are within a right angle, then they add up to 90 degrees.



Angles on a Straight Line

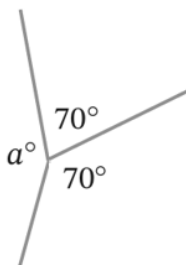
If angles are on a straight-line, then they add up to 180 degrees.



FOUNDATION

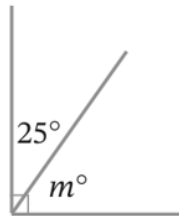
1 Calculate the value of the unknown angles. You do not need to show reasoning.

a



220°

b



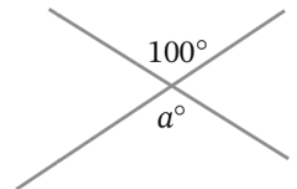
65°

c



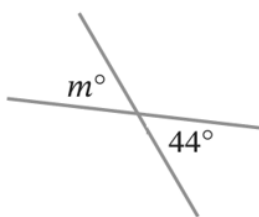
325°

d



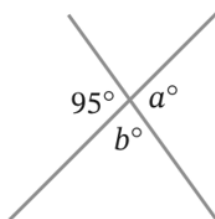
100°

e



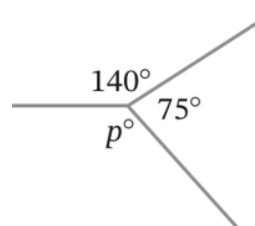
44°

f



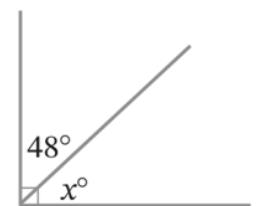
$a = 95^\circ, b = 85^\circ$

g



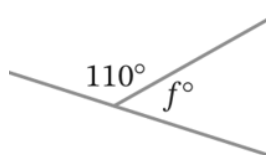
145°

h



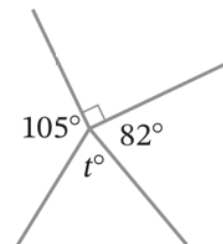
42°

i



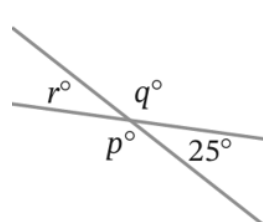
70°

j



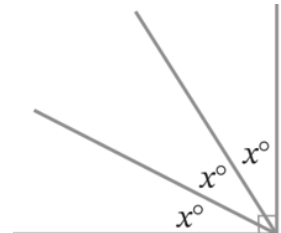
83°

k



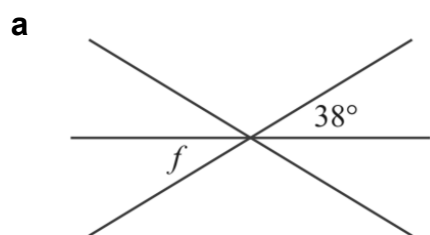
$r = 25^\circ, p = q = 155^\circ$

l

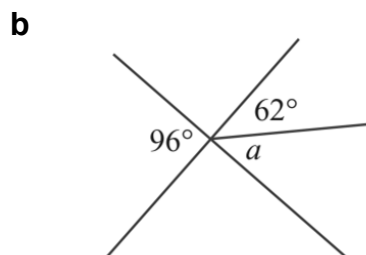


30°

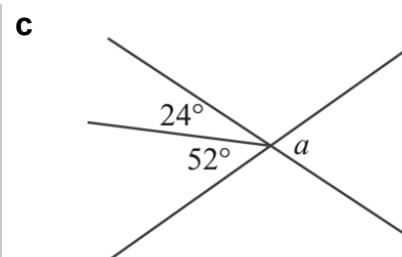
- 2 Combine your knowledge of angle relationships to calculate the value of each angle. You do not need to give reasoning.



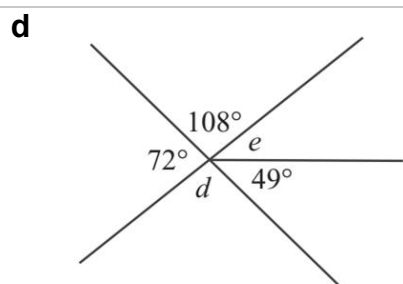
38°



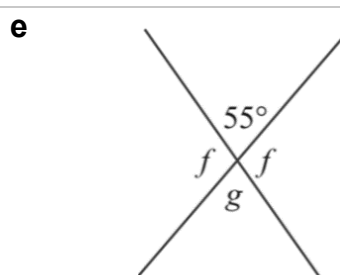
34°



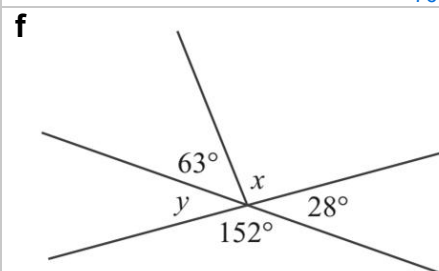
76°



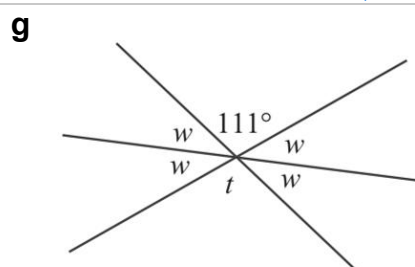
$d = 108^\circ, e = 23^\circ$



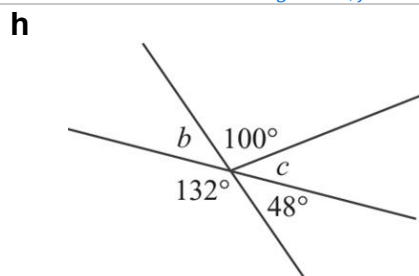
$g = 55^\circ, f = 125^\circ$



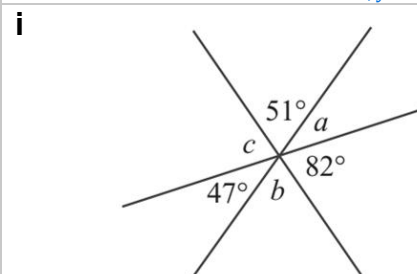
$x = 89^\circ, y = 28^\circ$



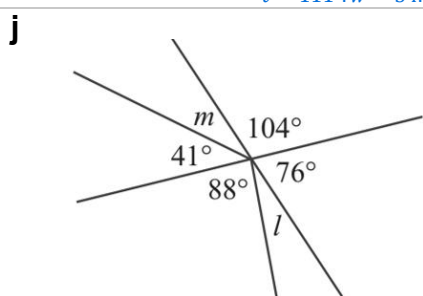
$t = 111^\circ, w = 34.5^\circ$



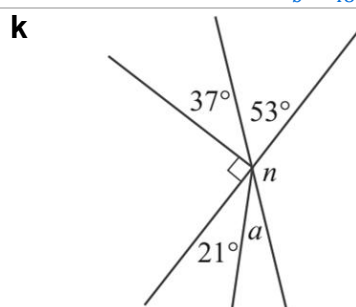
$b = 48^\circ, c = 32^\circ$



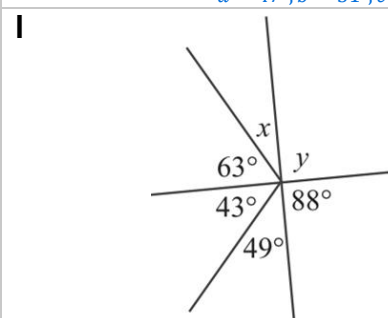
$a = 47^\circ, b = 51^\circ, c = 82^\circ$



$l = 16^\circ, m = 35^\circ$

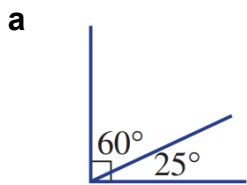


$a = 32^\circ, n = 127^\circ$



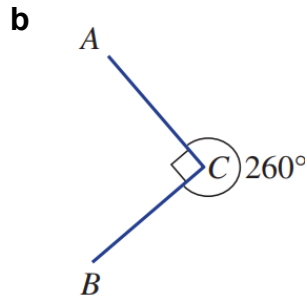
$x = 25^\circ, y = 92^\circ$

3 Decide whether these diagrams have the correct information or not. Justify your answer.

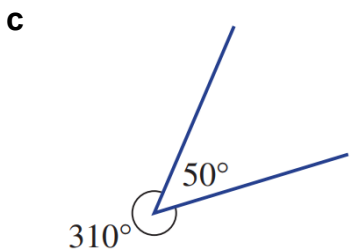


$60 + 25 = 85^\circ$

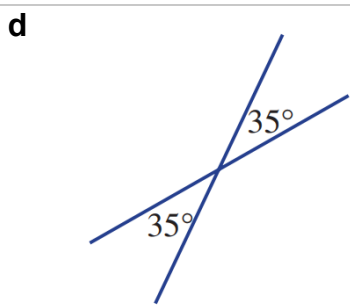
No, as angles in a right angle must sum to 90° .



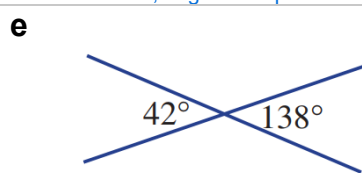
No, angles at a point sum to 360°



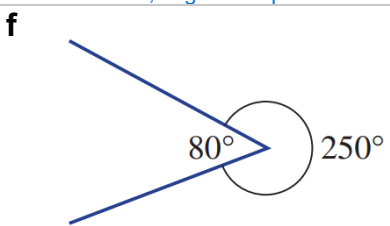
Yes, angles at a point sum to 360°



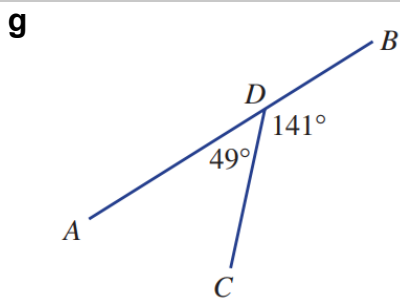
Yes, vertically opposite angles are equal.



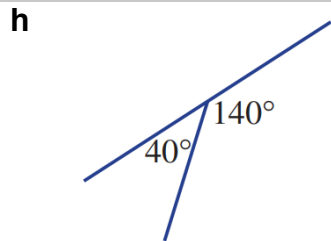
No, vertically opposite angles are equal.



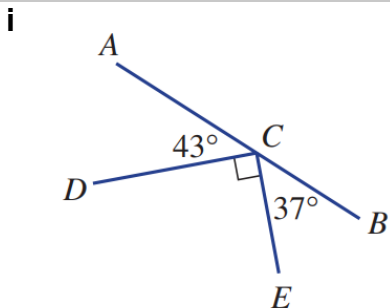
No, angles at a point sum to 360°



No, angles on a straight line sum to 180°

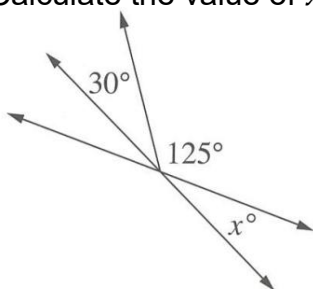


Yes, angles on a straight line sum to 180°



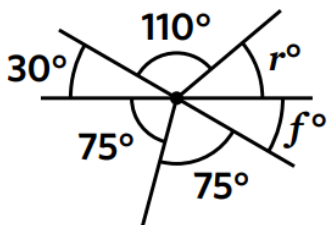
No, angles on a straight line sum to 180°

- 4 Calculate the value of x .



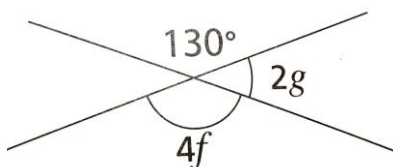
$$x = 25^\circ$$

- 5 Calculate the value of r and f



$$r = 40^\circ, f = 30^\circ$$

- 6 Which has the larger value, f or g ? Explain your reasoning.



$$f = 32,5, g = 25, \therefore f > g$$

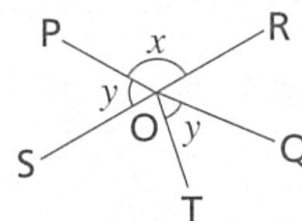
- 7 Write expressions in terms of x and/or y for

a $\angle ROQ$

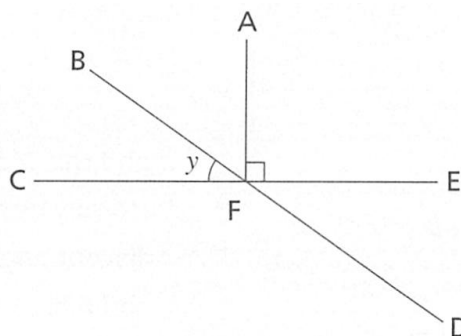
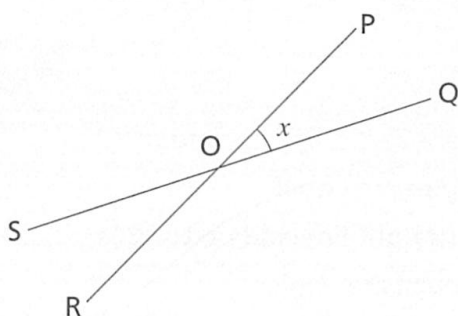
$$\angle ROQ = y$$

b $\angle SOT$

$$\angle SOT = x - y$$



- 8 Consider the two diagrams.



Write an expression for:

a $\angle ROS$

b $\angle POS$

c $\angle ROQ$

d $\angle DFC$

e $\angle EFD$

f $\angle AFB$

$$x$$

$$180 - x$$

$$180 - x$$

$$180 - y$$

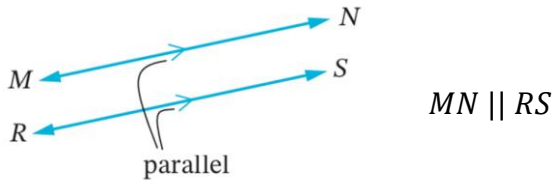
$$y$$

$$90 - y$$

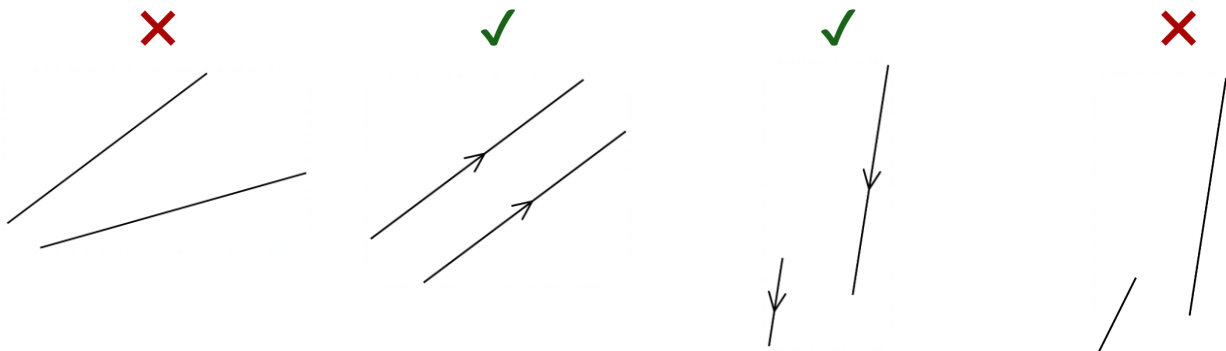
Perpendicular and Parallel Lines

i Parallel Lines

Parallel lines run in the same direction and never meet, even if extended forever. Diagrams show them with arrows $>$; statements use the $||$ symbol.

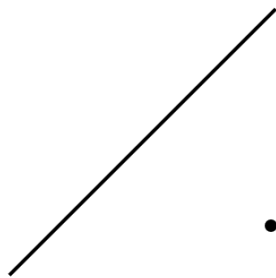


? Identify parallel lines



Draw a parallel line that passes through the given point.

a



b



c



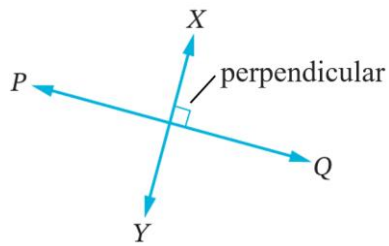
d



i Perpendicular Lines

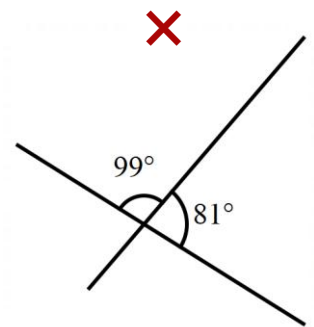
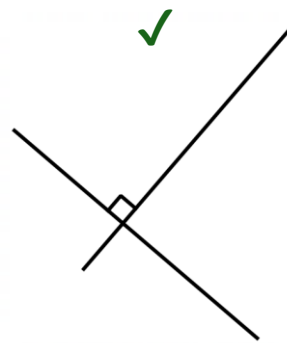
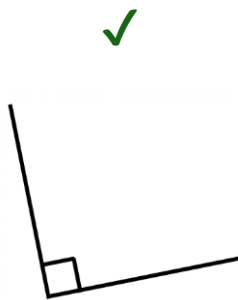
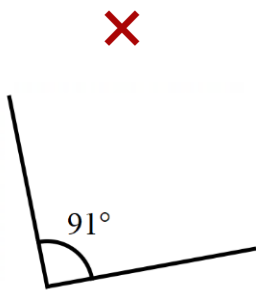
Perpendicular lines intersect at a **right angle**.

Diagrams show this with a square; statements use the $\perp$ symbol.



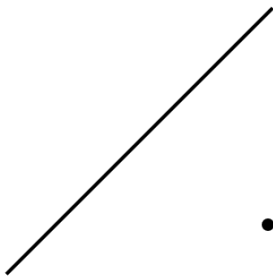
$$PQ \perp XY$$

ii Identify perpendicular lines



Draw a perpendicular line that passes through the given point.

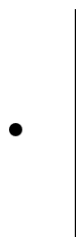
a



b



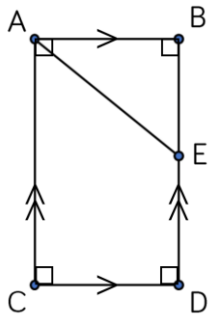
c



d



Example Identify perpendicular and parallel lines



Name two pairs of parallel lines in the diagram

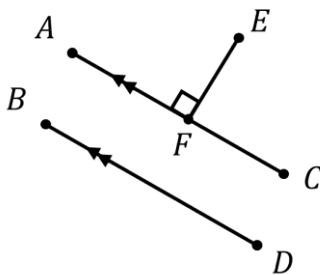
$AB \parallel CD$ and $AC \parallel BD$

Name two pairs of perpendicular lines in the diagram.

$AC \perp CD$ and $BD \perp AB$

Guided Practice

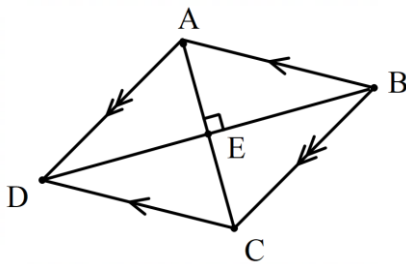
a



Name a pair of parallel lines.

Name a pair of perpendicular lines.

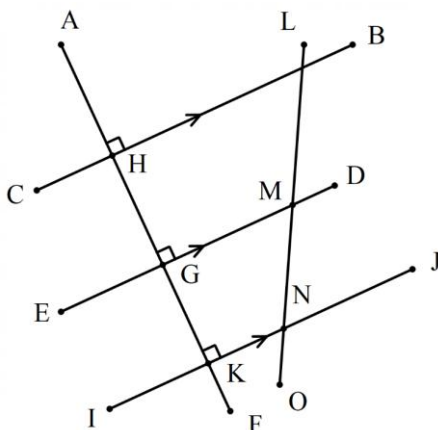
b



Name two pairs of parallel lines.

Name a pair of perpendicular lines.

c



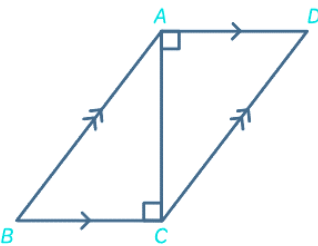
Name a set of three parallel lines.

Which line is perpendicular to all three parallel lines?

Key Ideas

- Two lines that are in the same direction are called
- Two lines that meet at 90° are called

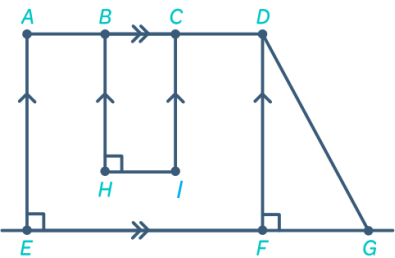
1 Complete the statements:



- a DA is parallel to
b AB is to CD
c AC is perpendicular to and

CB
parallel
BC and AD

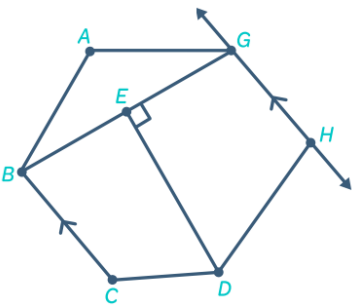
2 Consider the diagram of a swing set on the right.



- a How many line segments are parallel to AE?
b Identify 2 line segments perpendicular to EG.

3: BH, CI, DF
AE or DF

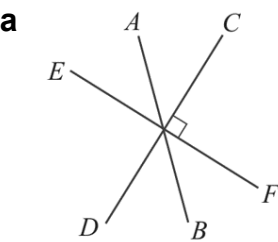
3 Consider the diagram to the right.



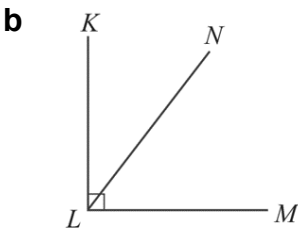
- a Name a line segment parallel to GH
b Identify a line segment perpendicular to GE

BC
DE

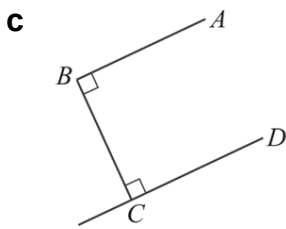
4 Name all the pairs of perpendicular lines in the following diagrams.
Write a statement like $WX \perp YZ$



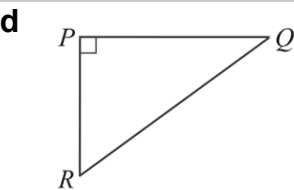
$CD \perp EF$



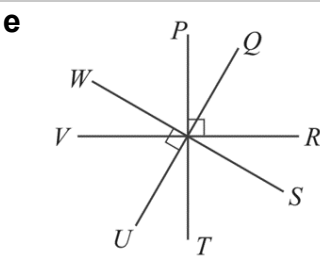
$KL \perp LM$



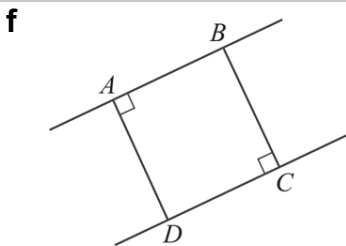
$AB \perp BC, BC \perp CD$



$RP \perp PQ$



$PT \perp PR, WS \perp UQ$

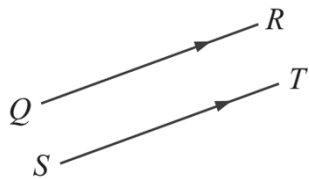


$AB \perp AD, BC \perp DC$

5 Name the pairs of parallel lines in these diagrams.

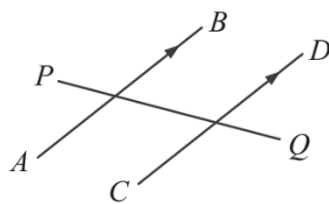
Write a statement like $WX \parallel YZ$

a



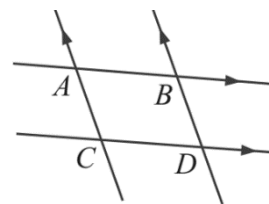
$QR \parallel ST$

b



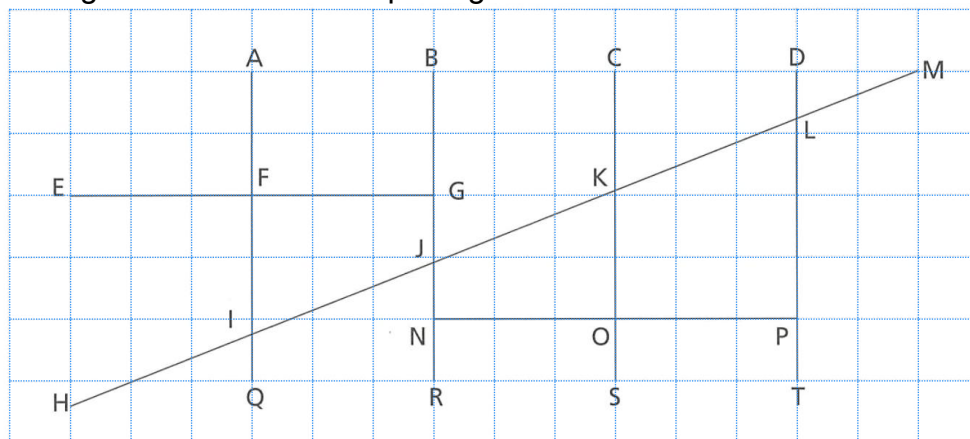
$AB \parallel CD$

c



$AB \parallel CD, AC \parallel BD$

6 A diagram is drawn on a square grid.



Decide whether these statements are true or false.

a AQ is parallel to BR.

true

b NP is perpendicular to CS.

true

c BR is perpendicular to HM.

false

d BR is parallel to DT.

true

e Add arrows and right-angle symbols to show parallel and perpendicular lines.

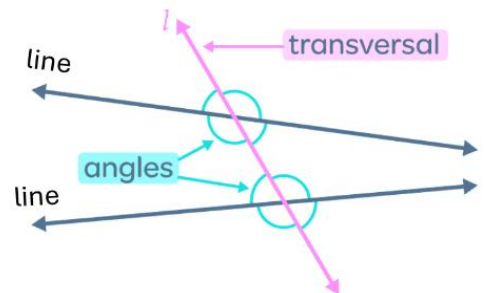
Angles on Transversals

i Transversals

A **transversal** is a line that intersects two or more lines.

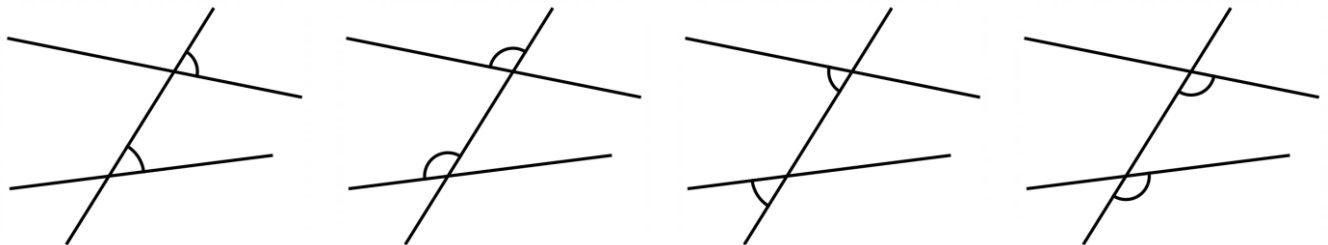
A transversal forms **three types** of angle pairs:

- **Corresponding** angles
- **Alternate** angles
- **Co-interior** angles



i Corresponding Pairs

Corresponding angle pairs have matching positions on the same side of the transversal.



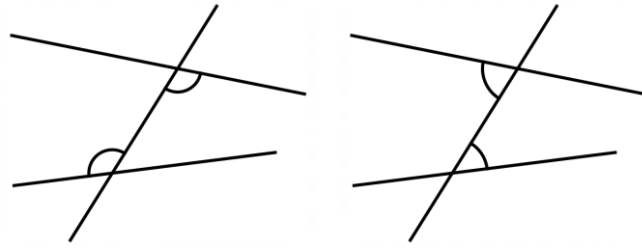
? Identify the corresponding angle

Draw the angle that is corresponding to the one shown on the diagram.

a 	b 	c
d 	e 	f

i Alternate Pairs

Alternate angle pairs are on opposite sides of the transversal, inside the lines.



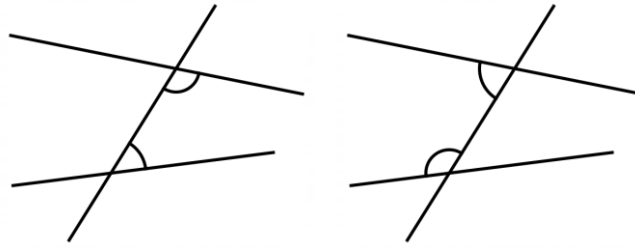
? Identify the alternate angle

Draw the angle that is alternate to the one shown on the diagram.

a 	b 	c
d 	e 	f

i Co-interior Pairs

Co-interior angle pairs are on the same side of the transversal, inside the lines.



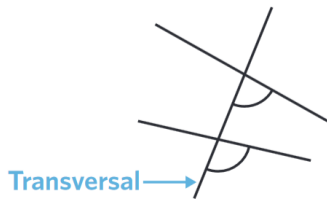
? Identify the co-interior angle

Draw the angle that is co-interior to the one shown on the diagram.

a 	b 	c
d 	e 	f

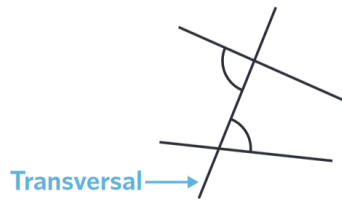
i Angles on Transversals

Corresponding Pairs



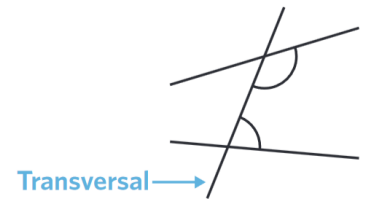
Matching positions on same side of transversal.

Alternate Pairs



Opposite sides of transversal, inside the lines.

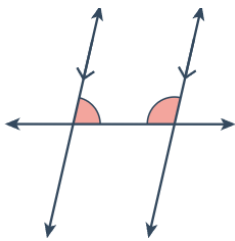
Co-interior Pairs



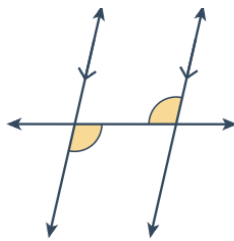
Same side of transversal, inside the lines.

? Identify transversal angle relationships

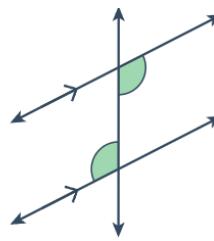
a



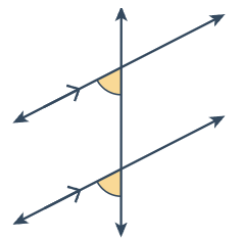
b



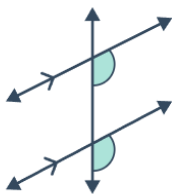
c



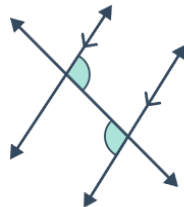
d



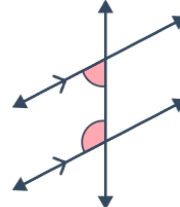
e



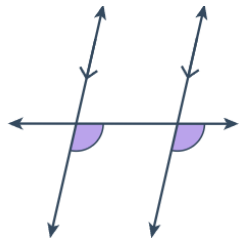
f



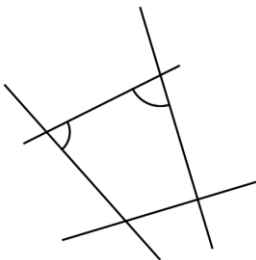
g



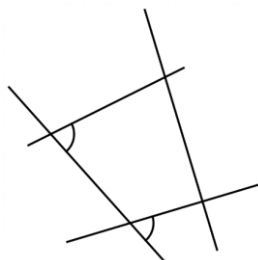
h



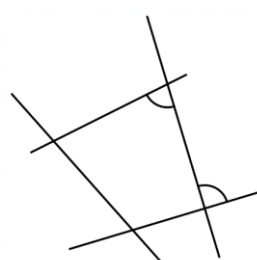
i



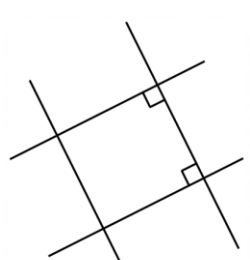
j



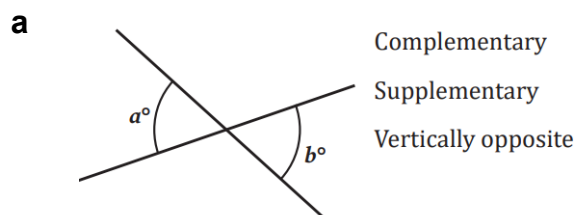
k



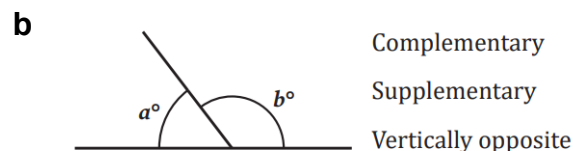
l



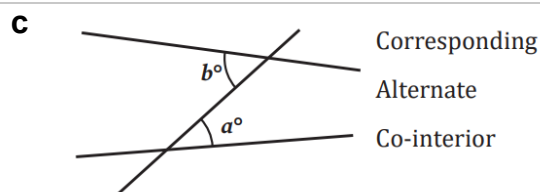
1 Identify the angle relationship shown.



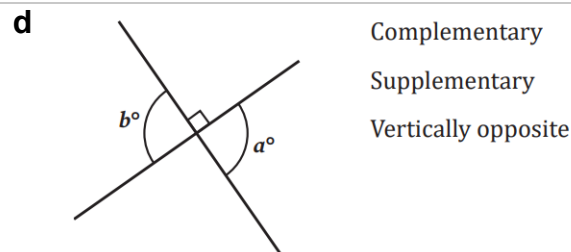
vertically opposite



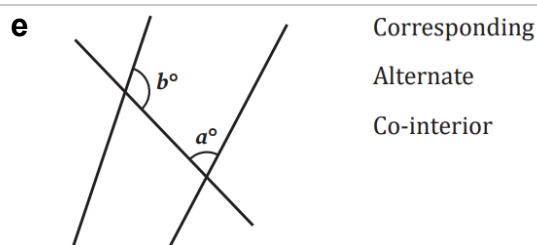
Supplementary



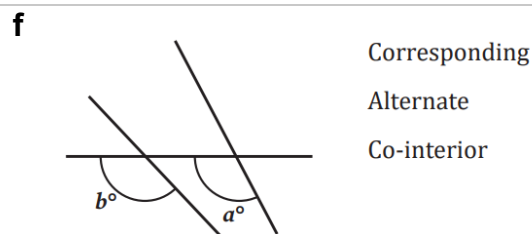
alternate



vertically opposite

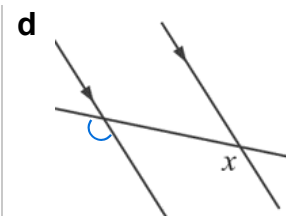
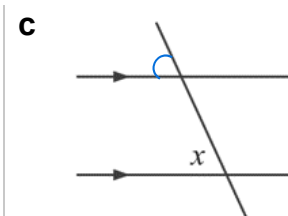
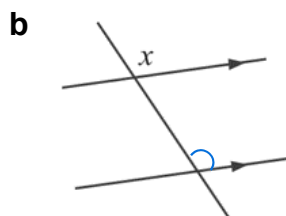
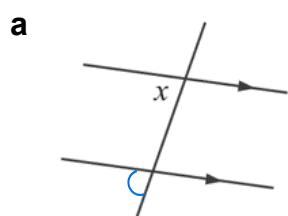


co-interior

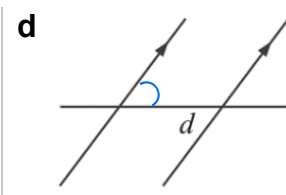
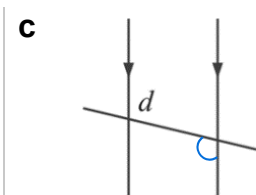
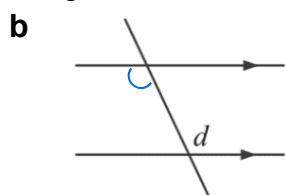
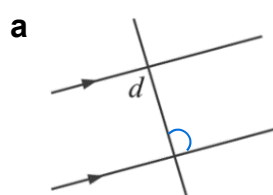


corresponding

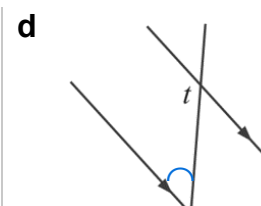
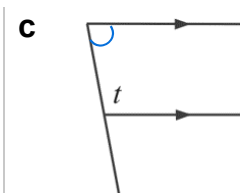
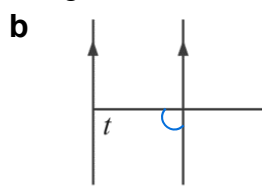
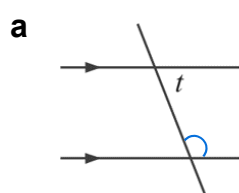
2 Draw an arc showing the angle corresponding to x .



3 Draw an arc showing the angle alternate to d .

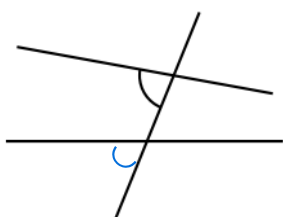


4 Draw an arc showing the angle co-interior to t .

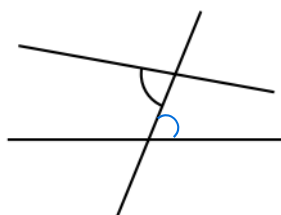


- 5** Label the missing angle arc for the corresponding, alternate, or co-interior angle.
There can be more than one answer, or no answer.

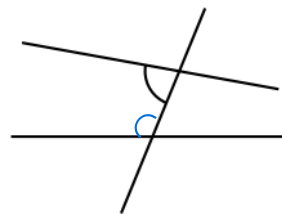
a Corresponding



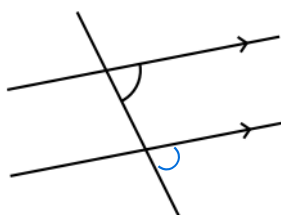
b Alternate



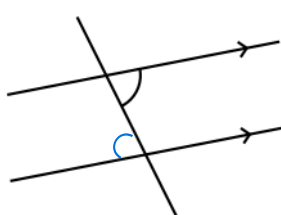
c Co-interior



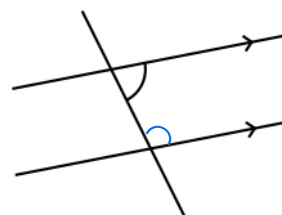
d Corresponding



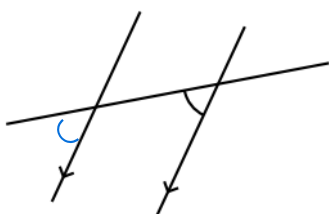
e Alternate



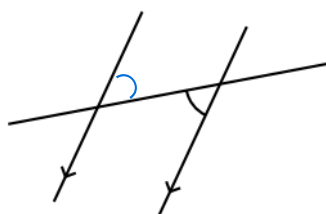
f Co-interior



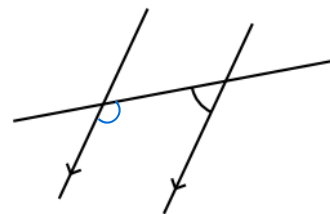
g Corresponding



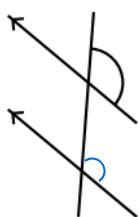
h Alternate



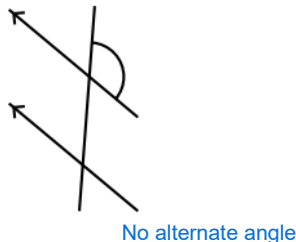
i Co-interior



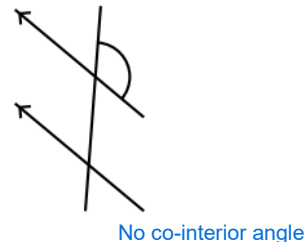
j Corresponding



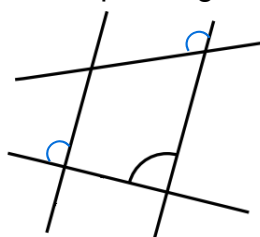
k Alternate



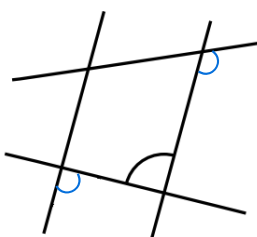
l Co-interior



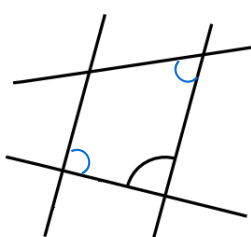
m Corresponding



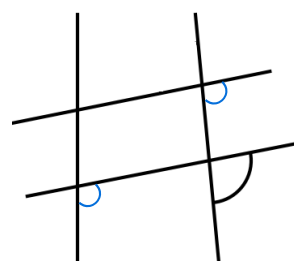
n Alternate



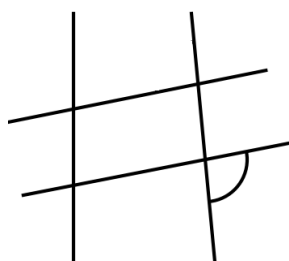
o Co-interior



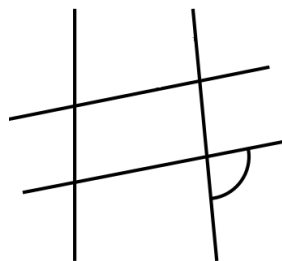
p Corresponding



q Alternate

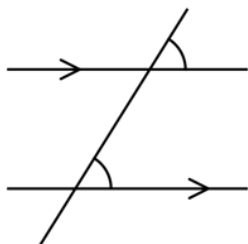


r Co-interior



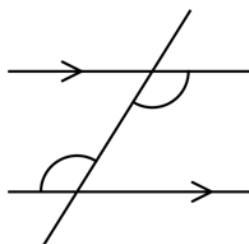
6 Identify whether these angles are corresponding, alternate, or co-interior.

a



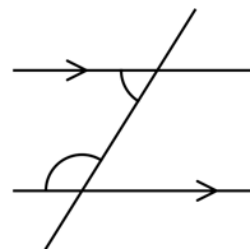
corresponding

b



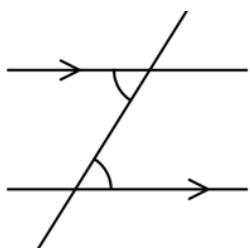
alternate

c



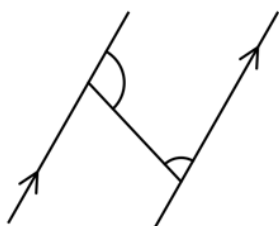
co-interior

d



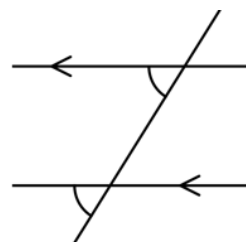
alternate

e



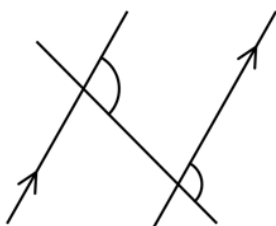
co-interior

f



corresponding

g



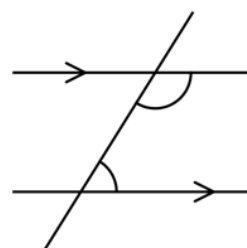
corresponding

h



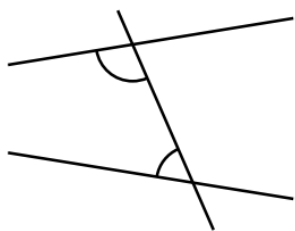
co-interior

i



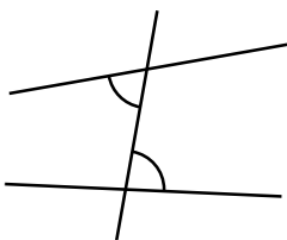
co-interior

j



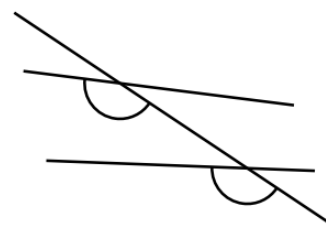
co-interior

k



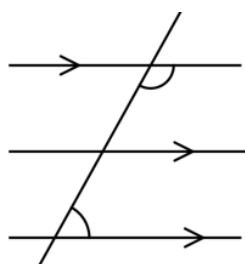
alternate

l



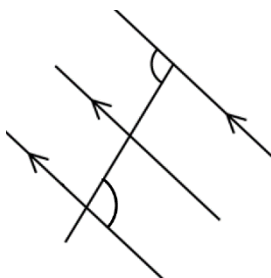
corresponding

m



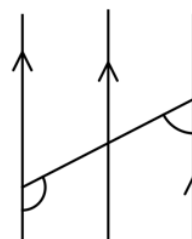
co-interior

n



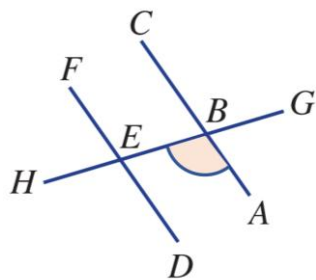
alternate

o



co-interior

7 Referring to the angle indicated, name the angle that is:



a Vertically opposite $\angle CBG$

b Corresponding

c Alternate

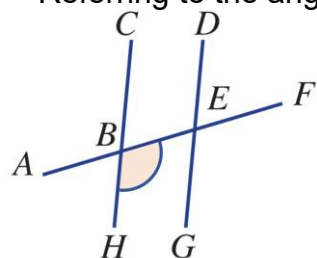
d Co-interior

$\angle HED$

$\angle FEB$

$\angle BED$

8 Referring to the angle indicated, name the angle that is:



a Vertically opposite

b Corresponding

c Alternate

d Co-interior

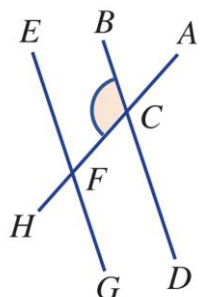
$\angle ABC$

$\angle GEF$

$\angle DEA$

$\angle GEB$

9 Referring to the angle indicated, name the angle that is:



a Vertically opposite

b Corresponding

c Alternate

d Co-interior

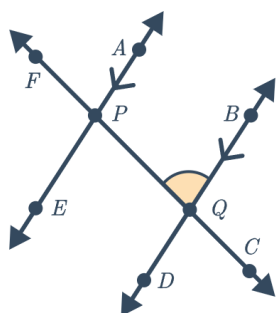
$\angle ACD$

$\angle EFH$

$\angle GFA$

$\angle EFA$

10 Referring to the angle indicated, name the angle that is:



a Vertically opposite

b Corresponding

c Alternate

d Co-interior

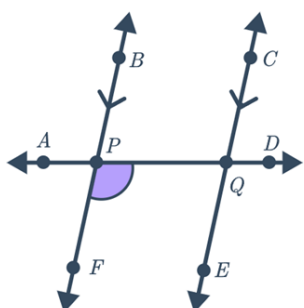
$\angle DQC$

$\angle FPA$

$\angle EPC$

$\angle APC$

11 Referring to the angle indicated, name the angle that is:



a Vertically opposite

b Corresponding

c Alternate

d Co-interior

$\angle APB$

$\angle EQD$

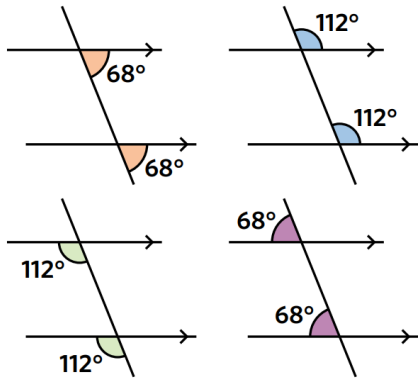
$\angle CQA$

$\angle EQA$

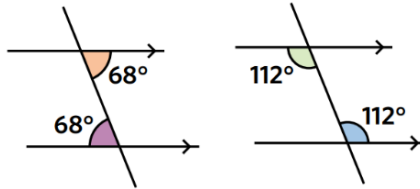
Angles on Parallel Lines

i Transversal Relationships on Parallel Lines

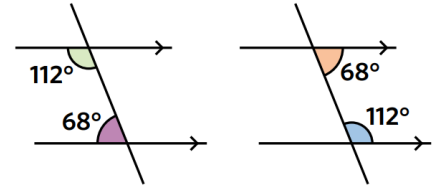
Corresponding angles on parallel lines are equal.



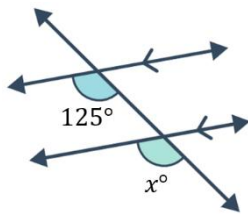
Alternate angles on parallel lines are equal.



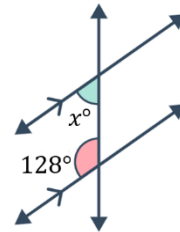
Co-interior angles between parallel lines sum to 180°



Example Calculate angles using parallel line properties



$$x = 125 \quad (\text{corresponding angles on } \parallel \text{ lines})$$

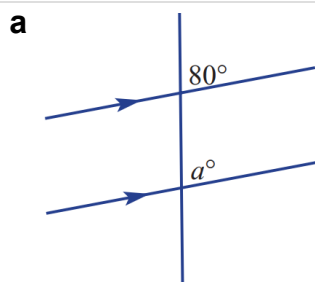


$$128 + x = 180 \quad (\text{co-interior angles in } \parallel \text{ lines})$$

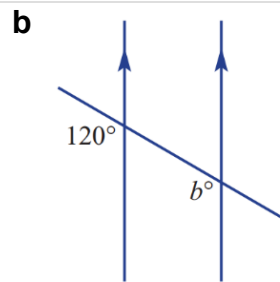
$$x = 52$$

Guided Practice

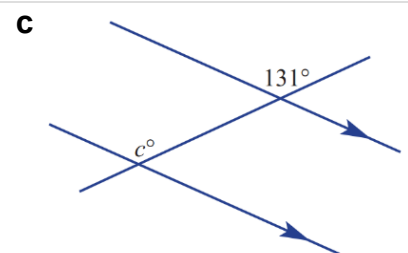
Find the value of the unknown angle. Give a brief reason.



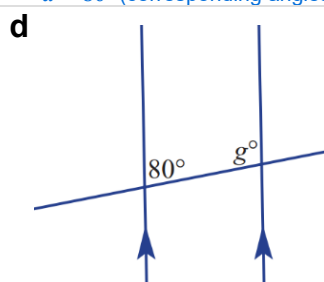
$$a = 80^\circ \quad (\text{corresponding angles on } \parallel \text{ lines})$$



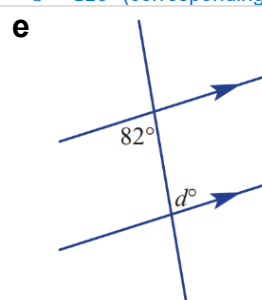
$$b = 120^\circ \quad (\text{corresponding angles on } \parallel \text{ lines})$$



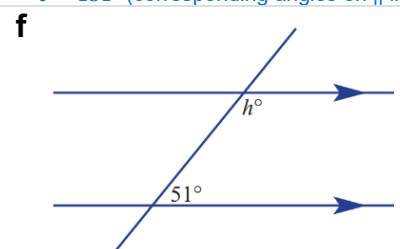
$$c = 131^\circ \quad (\text{corresponding angles on } \parallel \text{ lines})$$



$$g = 100^\circ \quad (\text{co-interior angles in } \parallel \text{ lines})$$



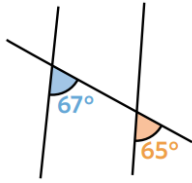
$$d = 82^\circ \quad (\text{alternate angles on } \parallel \text{ lines})$$



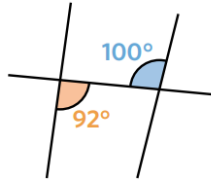
$$h = 129^\circ \quad (\text{co-interior angles on } \parallel \text{ lines})$$

i Deciding whether Two Lines are Parallel

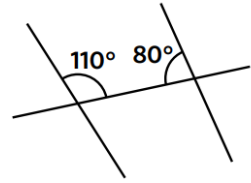
If corresponding angles are not equal, then the lines are not parallel.



If alternate angles are not equal, then the lines are not parallel.

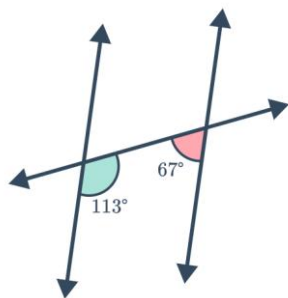


If co-interior angles do not sum to 180° , then the lines are not parallel.



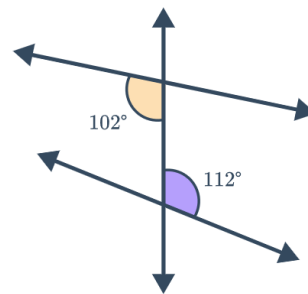
Example Determine whether two lines are parallel

Decide whether these lines are parallel.



The co-interior angles sum to 180°

Parallel.



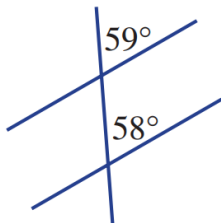
Alternate angles are not equal.

Not parallel.

Guided Practice

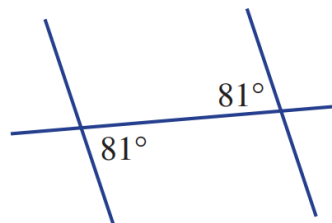
Decide whether these lines are parallel. Give a reason for your answer.

a



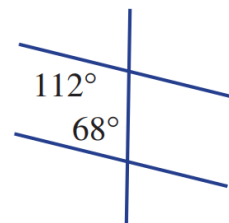
No, corresponding angles not equal.

b



Yes, corresponding angles are equal.

c

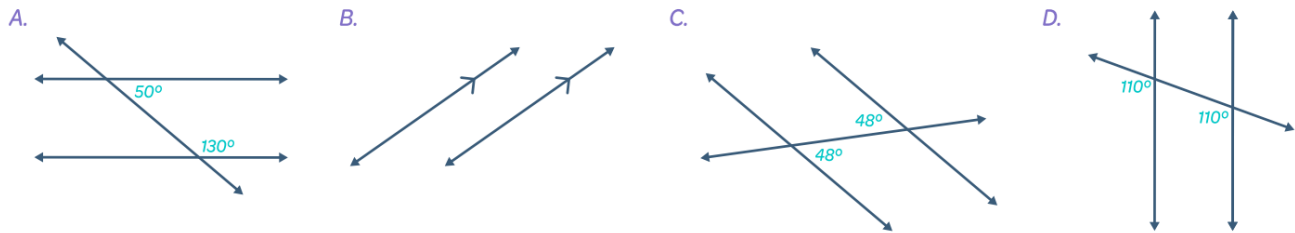


Yes, co-interior angles sum to 180° .

Key Ideas

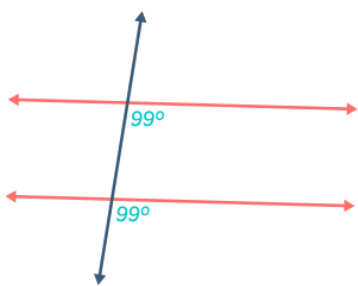
1. Corresponding angles on parallel lines are always
2. Alternate angles on parallel lines are always
3. Co-interior angles between parallel lines always

1 Match the diagram to the reason why the lines are parallel.



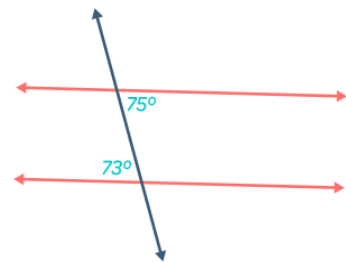
- Alternate angles are equal. C
- Corresponding angles are equal. D
- Co-interior angles add to 180°. A
- The diagram shows parallel line markings. B

2 Complete these statements.



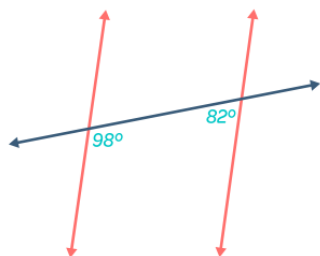
a The alternate | corresponding | co-interior angles are | aren't equal, so the lines are | aren't parallel.

The corresponding angles are equal, so the lines are parallel.



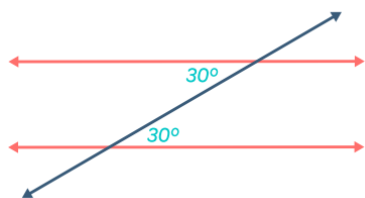
b The angles equal, so the lines parallel.

The alternate angles aren't equal, so the lines aren't parallel



c The angles add to 180, so the lines parallel.

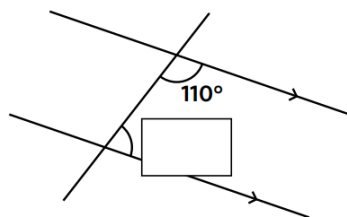
The co-interior angles don't add to 180, so the lines aren't parallel



d The angles equal, so the lines parallel.

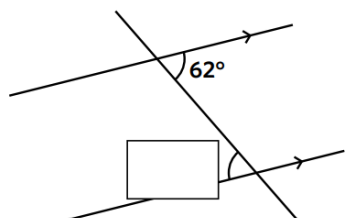
The alternate angles aren't equal, so the lines aren't parallel.

3 Calculate the value of the missing angle and give a reason.



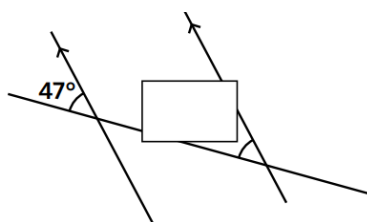
a angles
between parallel lines sum to 180°

70°, cointerior



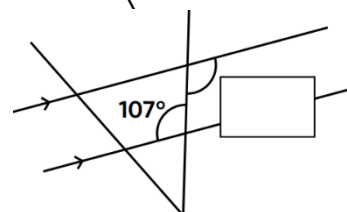
b angles
between parallel lines are equal.

62°, alternate



c angles
in parallel lines are equal.

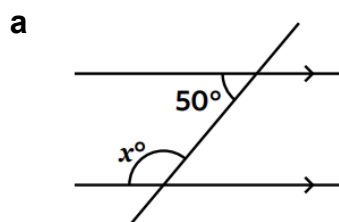
47°, corresponding



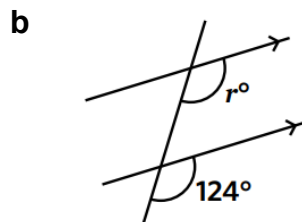
d angles
between parallel lines are equal.

107°, alternate

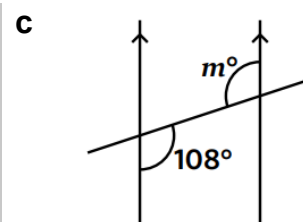
4 Calculate the value of the missing angle and give a reason.



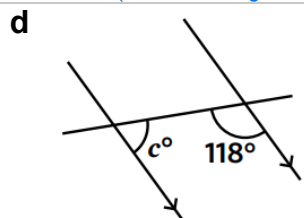
130° (co-interior angles in parallel lines)



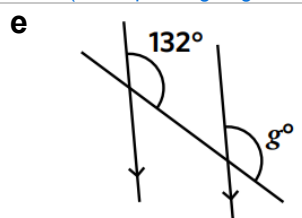
124° (corresponding angles on parallel lines)



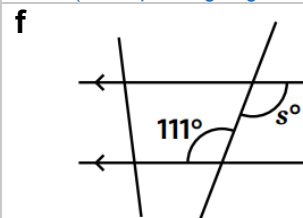
108° (corresponding angles on parallel lines)



62° (co-interior angles in parallel lines)



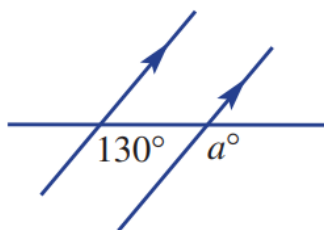
132° (corresponding angles on parallel lines)



111° (alternate angles on parallel lines)

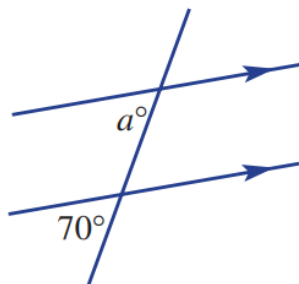
5 For each diagram, calculate the value of the missing angles. Give a brief reason.

a



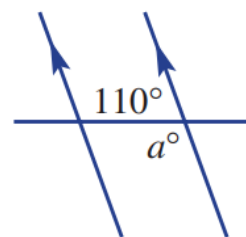
130 (corresponding angles on parallel lines)

b



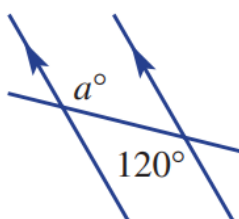
70 (corresponding angles on parallel lines)

c



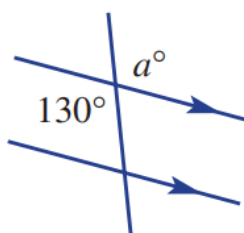
110 (alternate angles on parallel lines)

d



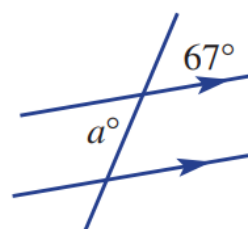
120 (alternate angles on parallel lines)

e



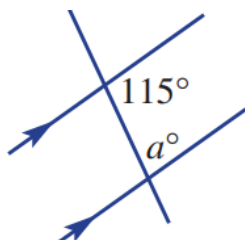
130 (vertically opposite angles)

f



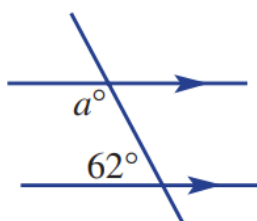
67 (vertically opposite)

g



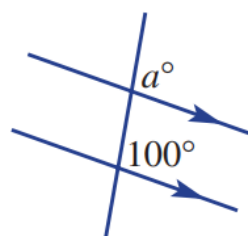
65 (co-interior angles between parallel lines)

h



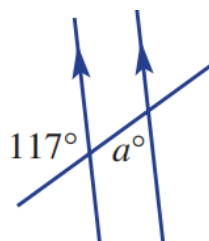
118 (co-interior angles between parallel lines)

i



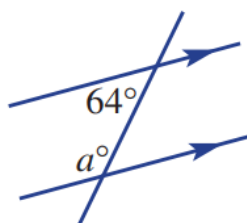
100 (corresponding angles on parallel lines)

j



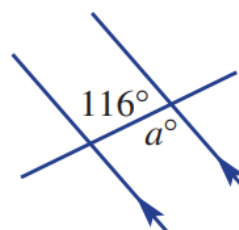
117 (vertically opposite angles)

k



116 (co-interior angles between parallel lines)

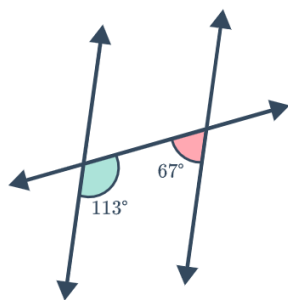
l



116 (alternate angles on parallel lines)

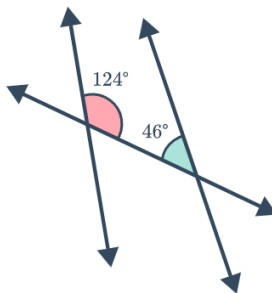
- 6 For each diagram, state whether the diagram has parallel lines. Justify your answer using angle relationships.

a



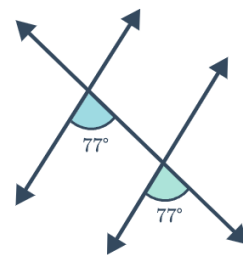
Yes. Co-interior angles sum to 180°

b



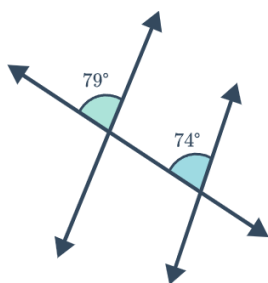
No. Co-interior angles don't sum to 180°

c



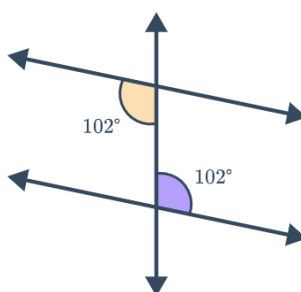
Yes. Corresponding angles are equal

d



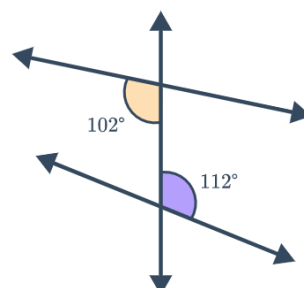
No. Corresponding angles aren't equal

e



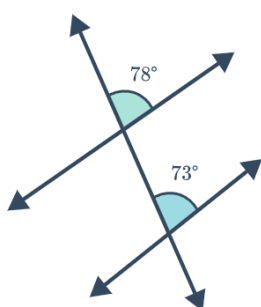
Yes. Alternate angles are equal

f



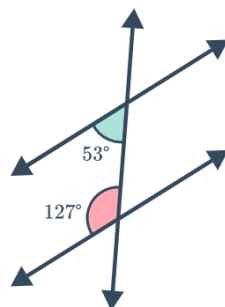
No. Alternate angles aren't equal

g



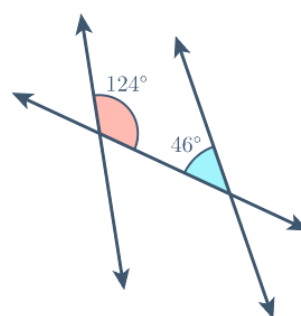
No. Corresponding angles aren't equal

h



Yes. Co-interior angles sum to 180°

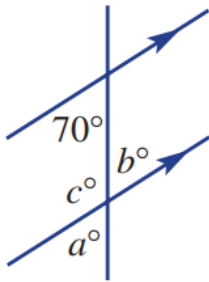
i



No. Co-interior angles don't sum to 180°

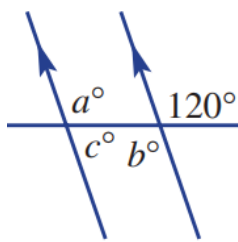
7 Calculate the value of each unknown angle in the diagram. You do not need to show reasoning.

a



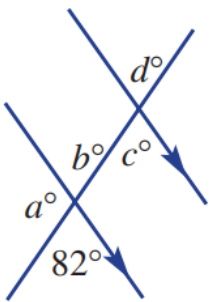
$a = 70$
 $c = 110$
 $b = 70$

b



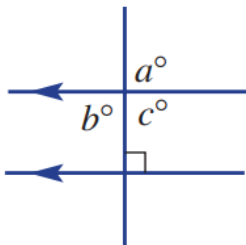
$a = 120$
 $b = 120$
 $c = 60$

c



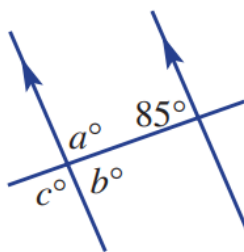
$a = 98$
 $b = 82$
 $c = 82$
 $d = 82$

d



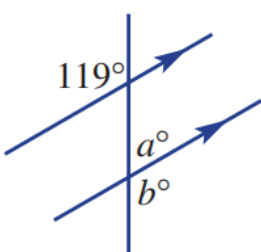
$a = 90$
 $b = 90$
 $c = 90$

e



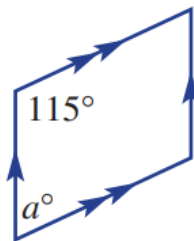
$a = 95$
 $b = 85$
 $c = 95$

f



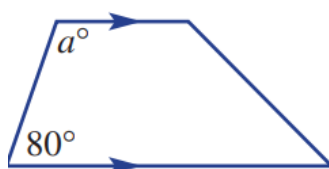
$b = 119$
 $a = 61$

g



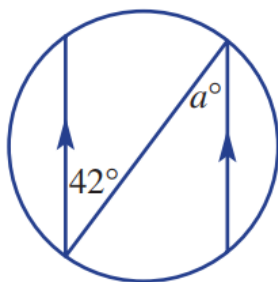
$a = 65$

h



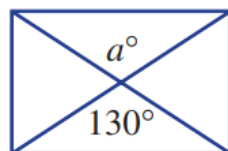
$a = 100$

i



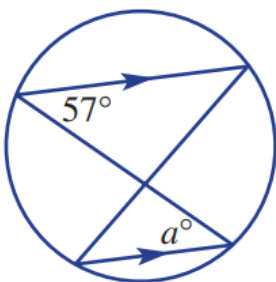
$a = 42$

j



$a = 130$

k

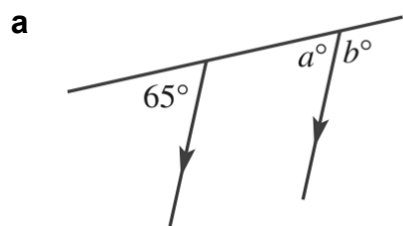


$a = 57$

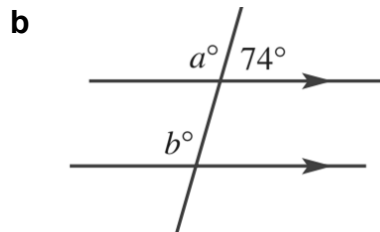
Problem-Solving involving Angles

FOUNDATION

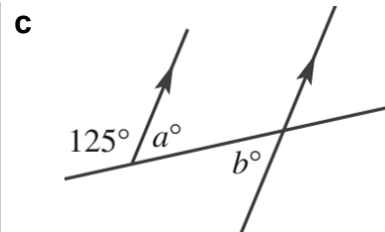
1 Calculate the values of the unknown angles in alphabetical order. Give a brief reason.



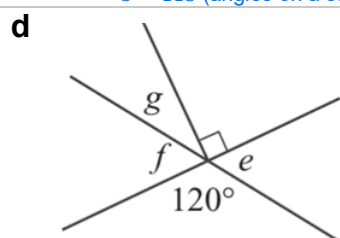
$a = 65$ (corresponding $\angle$ on $\parallel$ lines)
 $b = 115$ (angles on a straight line)



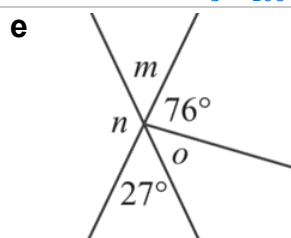
$a = 106$ (angles on a straight line)
 $b = 106$ (corresponding)



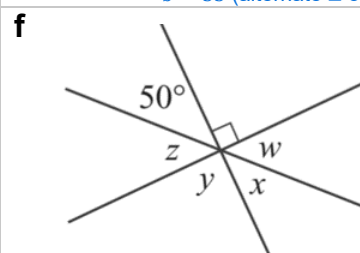
$a = 55$ (angles on a straight line)
 $b = 55$ (alternate $\angle$ on $\parallel$ lines)



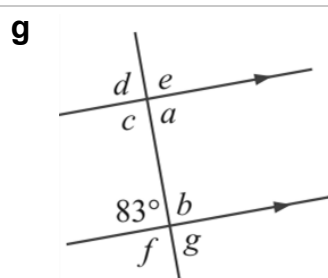
$e = 60$ (angles on a straight line)
 $f = 60$ (vertically opposite)
 $g = 30$ (angles in a right-angle)



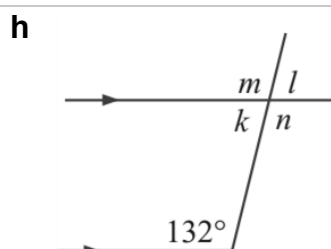
$m = 27$ (vertically opposite)
 $n = 153$ (angles on a straight line)
 $o = 77$ (angles on a straight line)



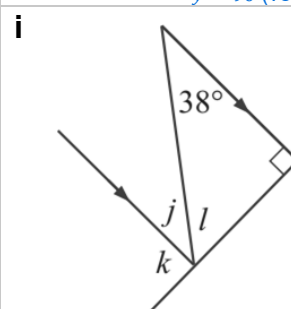
$x = 50$ (vertically opposite)
 $z = 40$ (angles on a straight line)
 $w = 40$ (vertically opposite)
 $y = 90$ (vertically opposite)



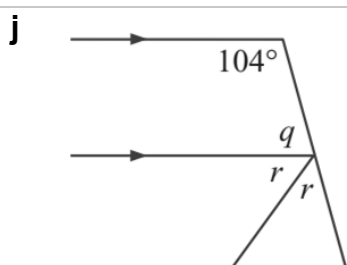
$a = 83$ (alternate $\angle$ on $\parallel$ lines)
 $b = 97$ (angles on a straight line)
 $c = 97$ (alternate to b)
 $d = 83$ (corresponding $\angle$ on $\parallel$ lines)
 $e = 97$ (vertically opposite to c)
 $f = 97$ (vertically opposite to b)
 $g = 83$ (vertically opposite)



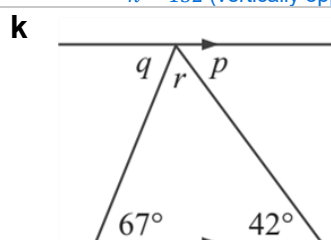
$k = 48$ (co-interior $\angle$ between $\parallel$ lines)
 $l = 48$ (vertically opposite to k)
 $m = 132$ (corresponding $\angle$ on $\parallel$ lines)
 $n = 132$ (vertically opposite to m)



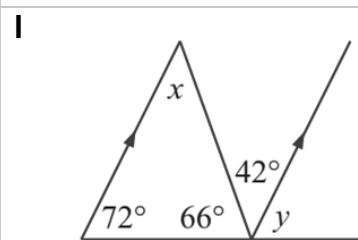
$j = 38$ (alternate $\angle$ on $\parallel$ lines)
 $l = 52$ (angles on a straight line)
 $k = 90$ (corresponding $\angle$ on $\parallel$ lines)



$q = 76$ (co-interior $\angle$ between $\parallel$ lines)
 $2r = 104$ (corresponding $\angle$ on $\parallel$ lines)
 $r = 52$



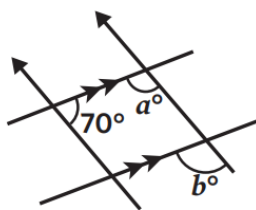
$q = 67$ (alternate $\angle$ on $\parallel$ lines)
 $p = 42$ (alternate $\angle$ on $\parallel$ lines)
 $r = 71$ (angles on a straight line)



$x = 42$ (alternate $\angle$ on $\parallel$ lines)
 $y = 72$ (angles on a straight line)

2 Find the value of the pronumerals. You do not need to give reasoning.

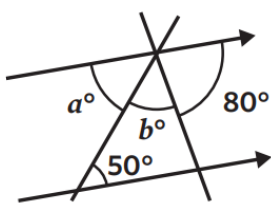
a



$$a = 110$$

$$b = 110$$

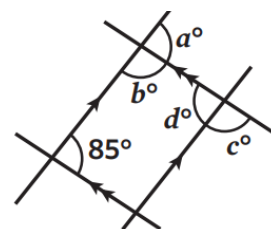
b



$$a = 50$$

$$b = 50$$

c



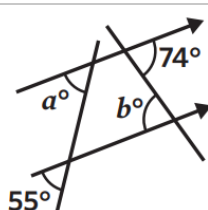
$$b = 95$$

$$a = 85$$

$$d = 85$$

$$c = 95$$

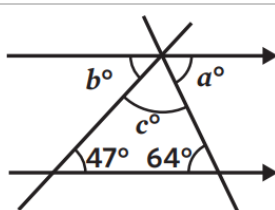
d



$$a = 55$$

$$b = 74$$

e

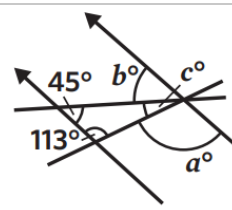


$$b = 47$$

$$a = 64$$

$$c = 69$$

f

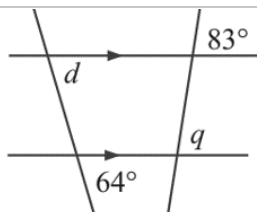


$$a = 113$$

$$b = 45$$

$$c = 22$$

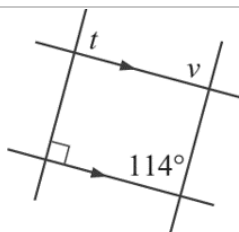
g



$$q = 83$$

$$d = 64$$

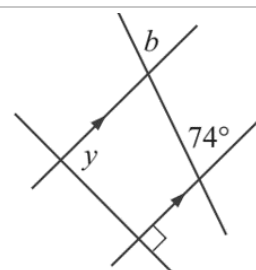
h



$$v = 114$$

$$t = 90$$

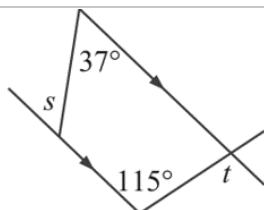
i



$$b = 74$$

$$y = 90$$

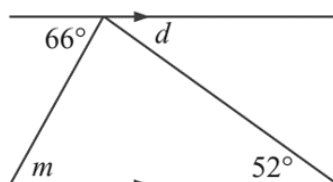
j



$$s = 37$$

$$t = 115$$

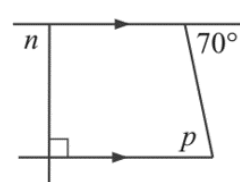
k



$$m = 66$$

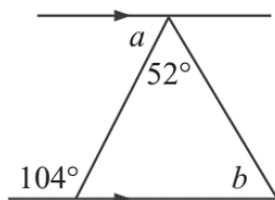
$$d = 52$$

l



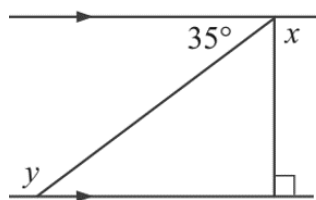
$$p = 70$$

$$n = 90$$

m

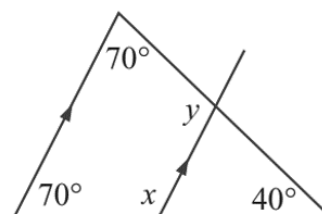
$$a = 76$$

$$b = 52$$

n

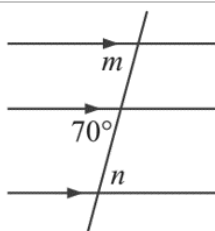
$$x = 90$$

$$y = 145$$

o

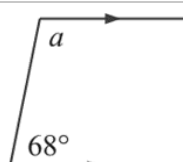
$$y = 110$$

$$x = 110$$

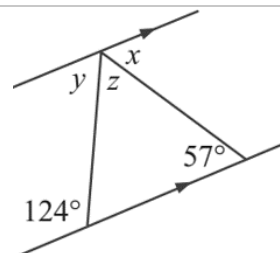
p

$$m = 70$$

$$n = 70$$

q

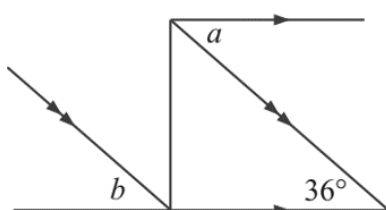
$$a = 112$$

r

$$x = 57$$

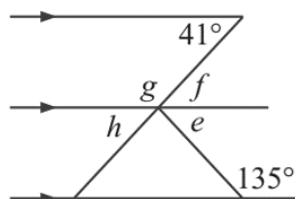
$$y = 56$$

$$z = 67$$

s

$$a = 36$$

$$b = 36$$

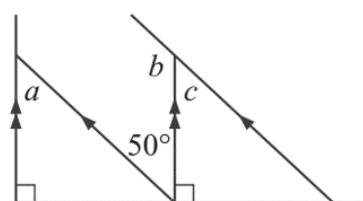
t

$$f = 41$$

$$g = 139$$

$$h = 41$$

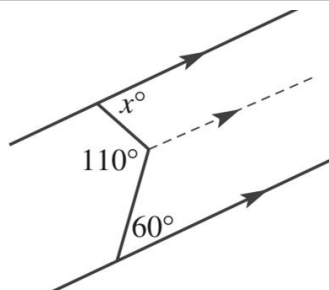
$$e = 45$$

u

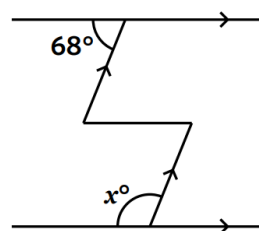
$$a = 50$$

$$b = 130$$

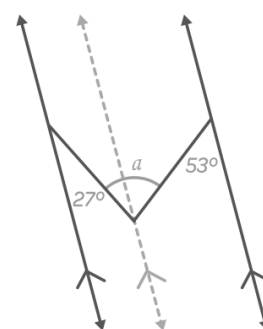
$$c = 50$$

v

$$x = 50$$

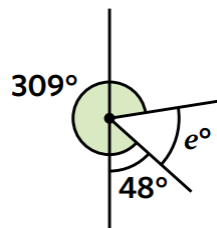
w

$$x = 112$$

x

$$a = 80$$

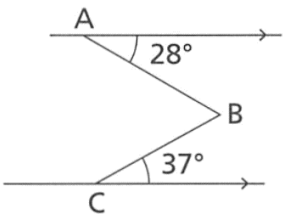
3 Find the value of e . Notice that 309° and 48° are overlapping.



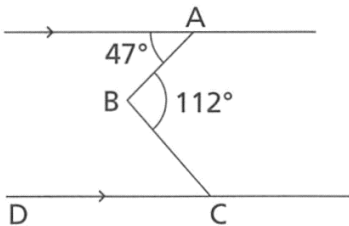
$e = 180 - 48 - 81 = 51$

4 Find these angles (hint: construct a parallel line)

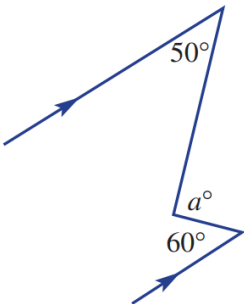
a Find the size of $\angle ABC$.



b Find the size of $\angle BCD$.



c

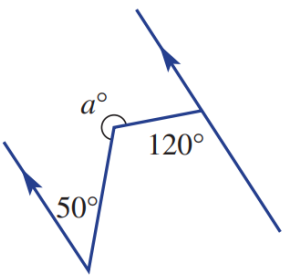


65°

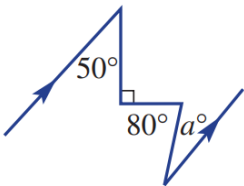
65°

110°

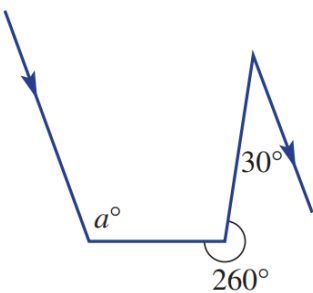
d



e



f

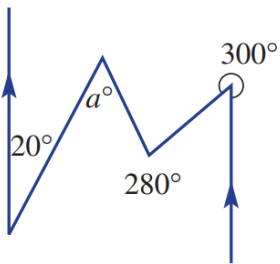


250°

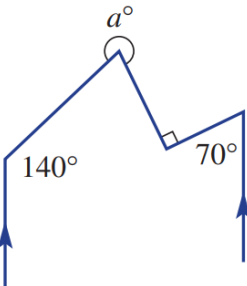
40°

110°

g



h

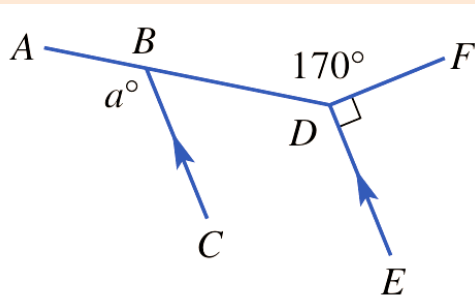


40°

300°

Writing Formal Reasoning

 **Example** Solve geometry problems and write formal reasoning



$$\angle BDE + 170 + 90 = 360 \text{ (angles at a point)}$$

$$\angle BDE + 260 = 360$$

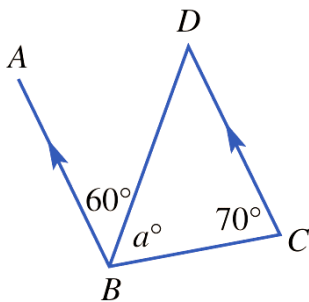
$$\angle BDE = 100$$

$$a = 100 \text{ (corresponding angles on } \parallel \text{ lines)}$$

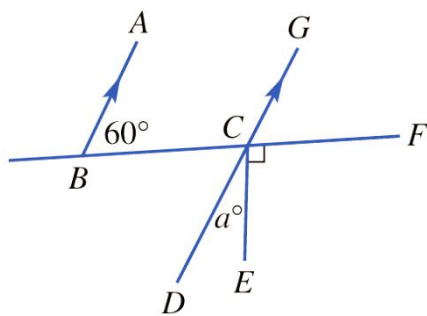
 **Guided Practice**

Find the value of angle a°

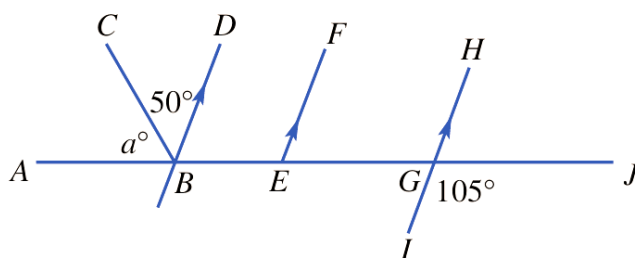
a



b

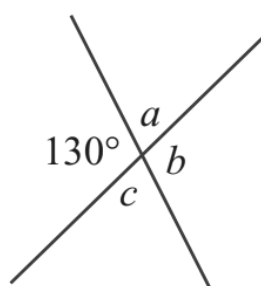


c



- 1 Complete the following to calculate a , b , c .

If you have a choice of reasons, use them to check your answers.

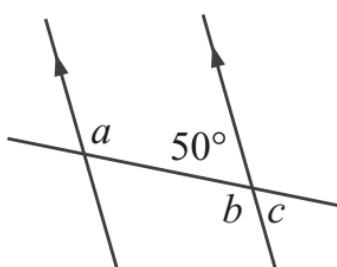


$$\begin{aligned} a + 130^\circ &= \underline{\hspace{2cm}}^\circ && \text{(straight angle)} \\ \therefore a &= \underline{\hspace{2cm}}^\circ \\ b &= 130^\circ && \text{(_____ opposite angles)} \\ c + 130^\circ &= 180^\circ && \text{(_____ angle)} \\ \therefore c &= \underline{\hspace{2cm}}^\circ \\ \text{or } c &= a = \underline{\hspace{2cm}}^\circ && \text{(_____ angles)} \end{aligned}$$

180 | 50 | vertically | straight | 50 | 50 | vertically opposite

- 2 Complete the following to calculate a , b , c .

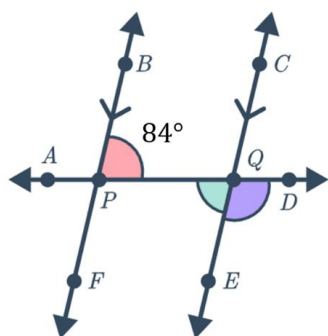
If you have a choice of reasons, use them to check your answers.



$$\begin{aligned} a + 50^\circ &= \underline{\hspace{2cm}}^\circ && \text{(co-interior angles on || lines)} \\ \therefore a &= \underline{\hspace{2cm}}^\circ \\ b + 50^\circ &= \underline{\hspace{2cm}}^\circ && \text{(straight angle)} \\ \therefore b &= \underline{\hspace{2cm}}^\circ \\ \text{or } b &= a = \underline{\hspace{2cm}}^\circ && \text{(_____ angles on || lines)} \\ c &= \underline{\hspace{2cm}}^\circ && \text{(_____ opposite angles)} \\ \text{or } c + b &= 180^\circ && \text{(_____ angle)} \\ \therefore c &= \underline{\hspace{2cm}}^\circ \end{aligned}$$

180 | 130 | 180 | 130 | 130 | alternate | 50 | vertically | straight | 50

- 3 Calculate $\angle AQE$ and $\angle DQE$ giving full reasoning.



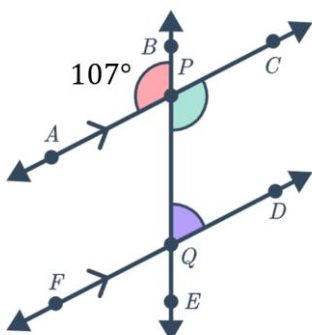
$$\begin{aligned} \angle AQE &= \underline{\hspace{2cm}}^\circ && \text{(..... angles on || lines)} \\ \angle DQE + \underline{\hspace{2cm}}^\circ &= 180^\circ && \text{(angles on a line)} \\ \therefore \angle DQE &= \underline{\hspace{2cm}}^\circ \end{aligned}$$

$\angle AQE = 84^\circ$ (alternate angles on || lines)

$\angle DQE + \angle AQE = 180^\circ$ (angles on a straight line)

$\therefore \angle DQE = 180 - 84 = 96^\circ$

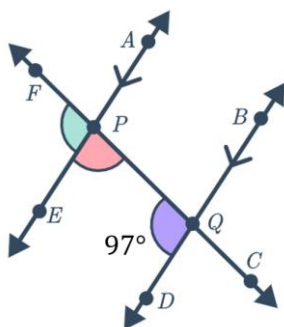
- 4 Calculate $\angle CPE$ and $\angle PQD$ giving full reasoning.



$\angle CPE = 107^\circ$ (vertically opposite)

$\angle PQD = 73^\circ$ (co-interior angles between parallel lines)

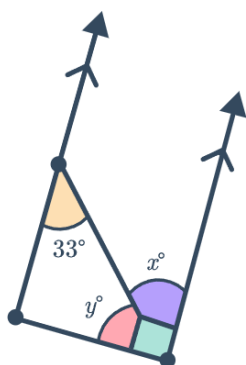
- 5 Calculate $\angle EPQ$ and $\angle EPF$ giving full reasoning.



$\angle EPQ = 83^\circ$ (co-interior angles between parallel lines)

$\angle FPE = 97^\circ$ (corresponding angles on parallel lines)

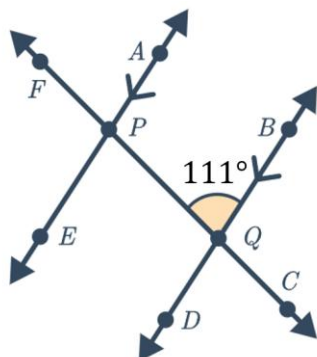
- 6 Calculate x and y giving full reasoning.



$x = 33^\circ$ (alternate angles on parallel lines)

$y = 57^\circ$ (angles in a right-angle)

- 7 Calculate $\angle FPE$ giving full reasoning.

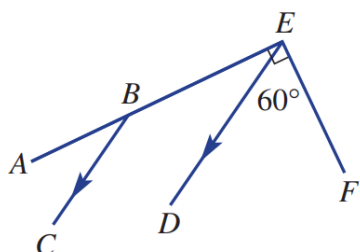


$\angle EPQ = 111^\circ$ (alternate angles on parallel lines)

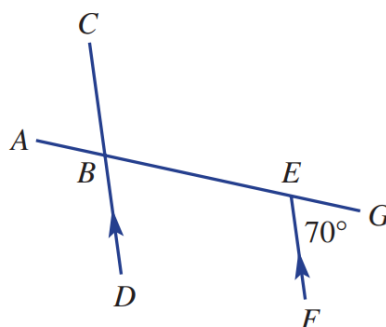
$\angle FPE = 69^\circ$ (angles on a straight line)

- 5 Calculate as many angles as you can in these diagrams. You do not need to show reasoning.

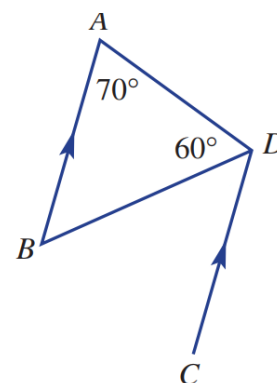
a

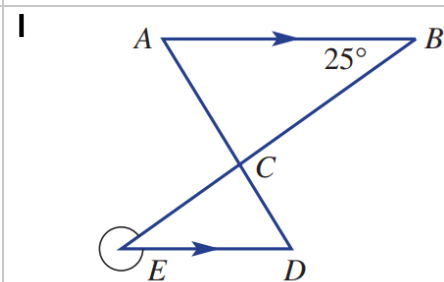
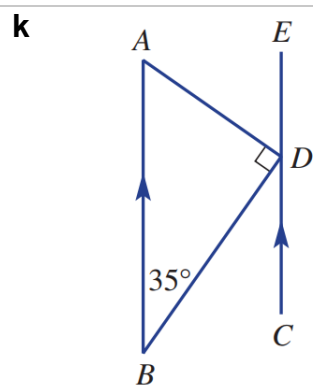
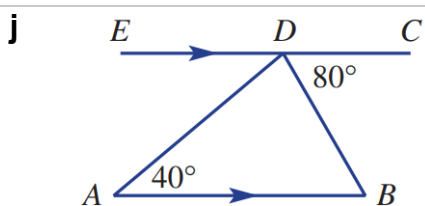
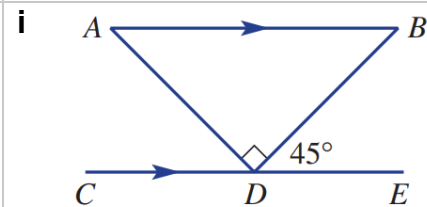
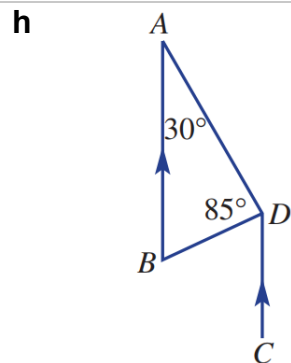
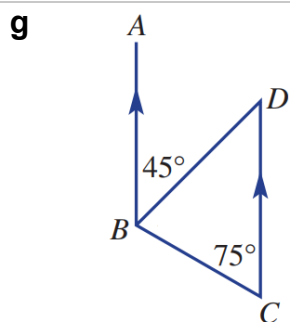
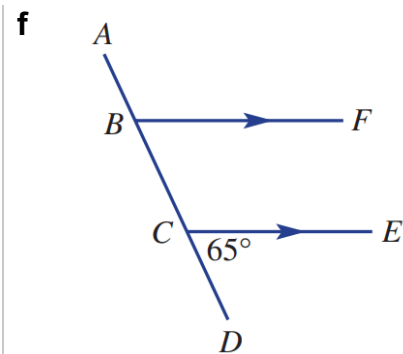
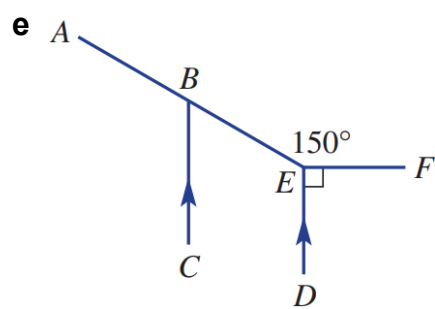
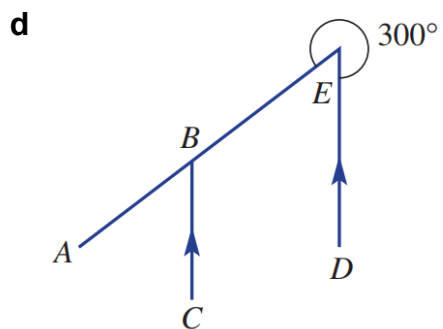


b



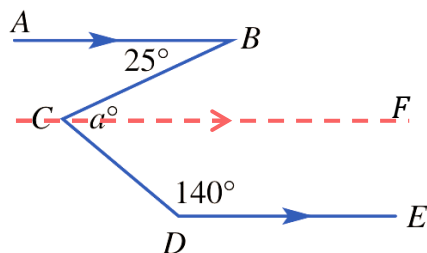
c





8 For these questions, you will need to construct a parallel line. Give full reasoning.

a



Construct CF parallel to AB and DE

$\angle BCF = \dots\dots^\circ$ (alternate angles on $\parallel$ lines)

$\angle FCD + 140 = 180^\circ$ (co-interior angles in $\parallel$ lines)

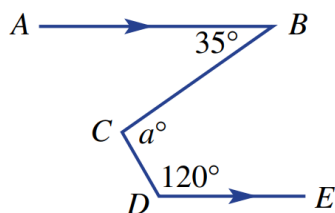
$\angle FCD = \dots\dots\dots$

$\therefore a = \dots\dots + \dots\dots$

$= \dots\dots\dots$

$$25^\circ + 40^\circ \mid 25^\circ + 40^\circ \mid a = 65^\circ$$

b



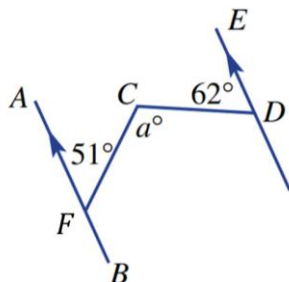
construct CF parallel to AB and DE

$\angle BCF = 35^\circ$ (alternate angles on $\parallel$ lines)

$\angle FCD = 60^\circ$ (co-interior angles in $\parallel$ lines)

$$a = 35 + 60 = 95^\circ$$

c



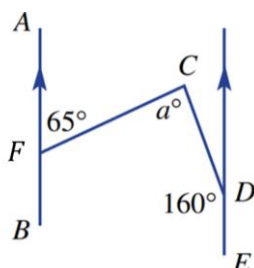
construct CG parallel to AB and DE

$\angle GCF = 51^\circ$ (alternate angles on $\parallel$ lines)

$\angle GCD = 62^\circ$ (alternate angles on $\parallel$ lines)

$$a = 51 + 62 = 113^\circ$$

d



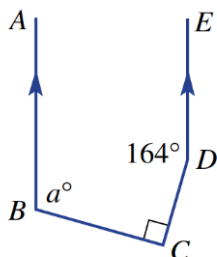
Construct CG parallel to AB and DE

$\angle GCF = 65^\circ$ (alternate angles on $\parallel$ lines)

$\angle GCD = 20^\circ$ (co-interior angles in $\parallel$ lines)

$$a = 65 + 20 = 85^\circ$$

e



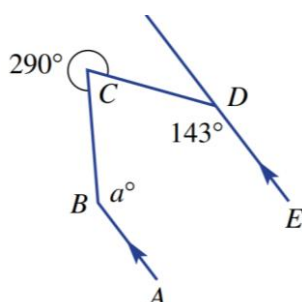
Construct CF parallel to AB and ED

$\angle DCF = 16^\circ$ (co-interior angles in $\parallel$ lines)

$\angle BCF = 74^\circ$ (angles in a right angle)

$a = 106^\circ$ (co-interior angles in $\parallel$ lines)

f



Construct CF parallel to AB and DE

$\angle DCF = 37^\circ$ (co-interior angles in $\parallel$ lines)

$\angle BCF = 33^\circ$ (angles at a point)

$a = 147^\circ$ (co-interior angles in $\parallel$ lines)

Geometry Problems involving Equations

FOUNDATION

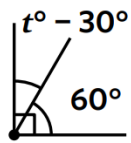
- 1 Use angle facts to form an equation for each diagram.
Then solve your equations to find the value of each letter.

a

$$t - 30 + 60 = 90$$

$$t + \dots = 90$$

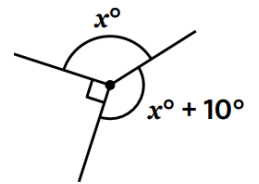
$$t = \dots$$



$$t = 60^\circ$$

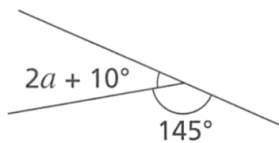
b

$$x + x + 10 + 90 = \dots$$



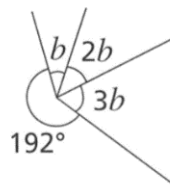
$$x = 130^\circ$$

c



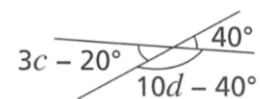
$$a = 12.5^\circ$$

d



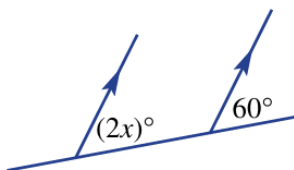
$$b = 28^\circ$$

e



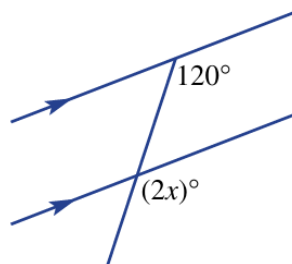
$$\begin{aligned} c &= 20^\circ \\ d &= 18^\circ \end{aligned}$$

f



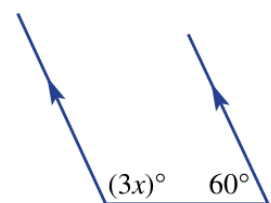
$$x = 30$$

g



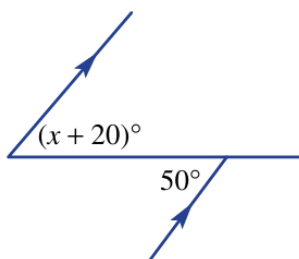
$$x = 60$$

h



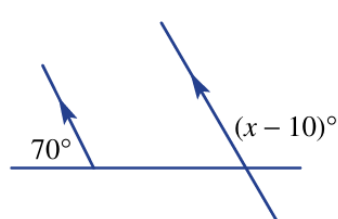
$$x = 40$$

i



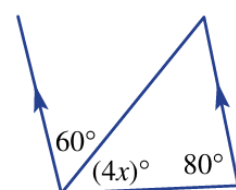
$$x = 30$$

j



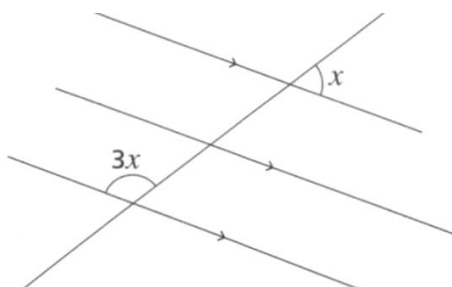
$$x = 120$$

k



$$x = 10$$

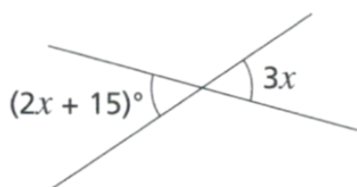
- 2 Work out the value of x .



$$x = 45^\circ$$

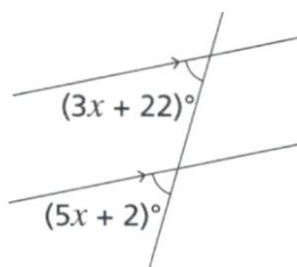
- 3 Work out value of x in these diagrams.

a



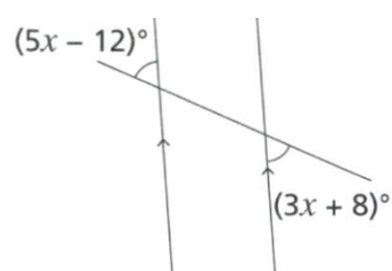
$$x = 15^\circ$$

b



$$x = 10^\circ$$

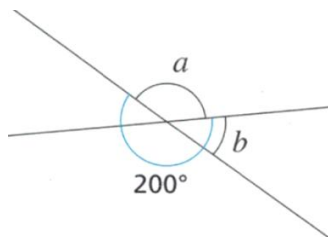
c



$$x = 10^\circ$$

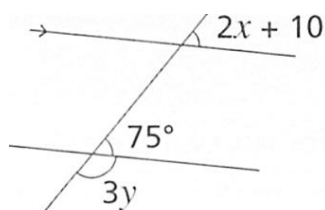
- 4 Determine the value of each pronumeral.

a



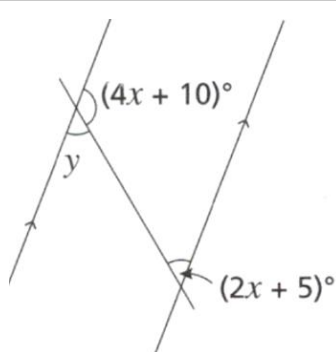
$$\begin{aligned} a &= 160^\circ \\ b &= 20^\circ \end{aligned}$$

b



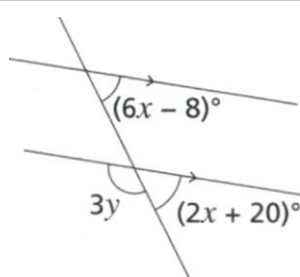
$$\begin{aligned} x &= 32.5^\circ \\ y &= 35^\circ \end{aligned}$$

c



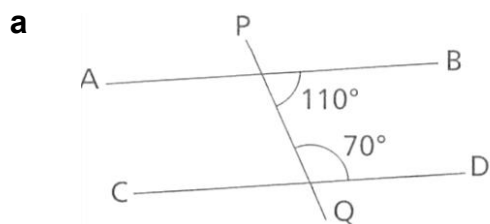
$$\begin{aligned} x &= 27.5^\circ \\ y &= 60^\circ \end{aligned}$$

d

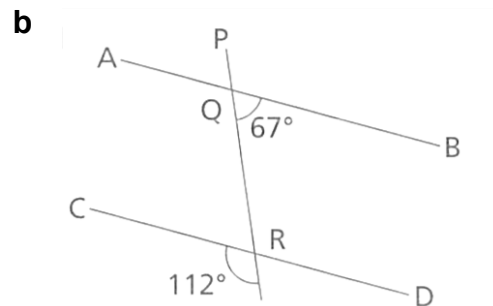


$$\begin{aligned} x &= 7^\circ \\ y &= 48\frac{2}{3}^\circ \end{aligned}$$

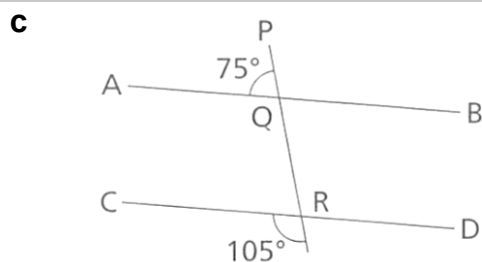
5 In each case, decide if AB is parallel to CD. Give a reason why or why not.



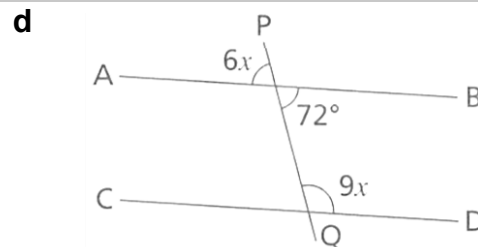
Yes, co-interior angles sum to 180.



$\angle PRD = 112^\circ$ (vertically opposite)
 $\therefore$ no, as co-interior angles don't sum to 180

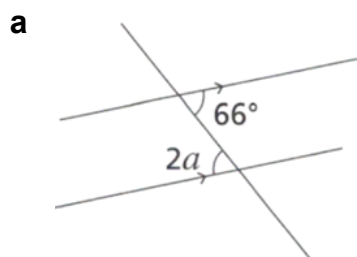


$\angle AQR = 105^\circ$ (angles on a straight line)
 $\therefore$ yes as corresponding angles are equal.

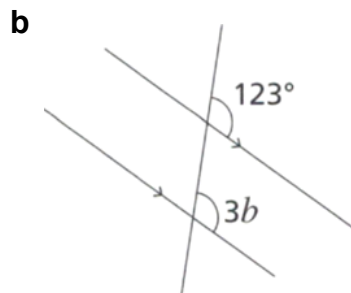


$6x = 72$
 $x = 12$
 $\angle PQD = 9x = 108^\circ$
 $\therefore$ yes as co-interior angles sum to 180

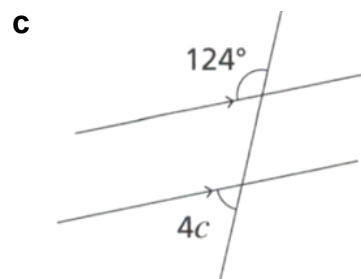
6 Work out the value of the pronumerals in each of these diagrams.



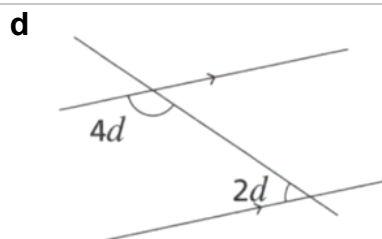
$$a = 33^\circ$$



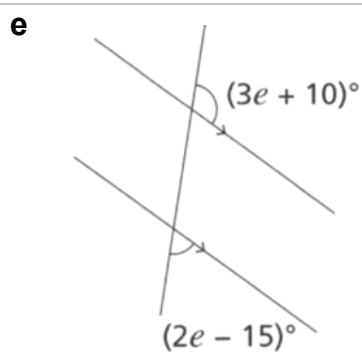
$$b = 41^\circ$$



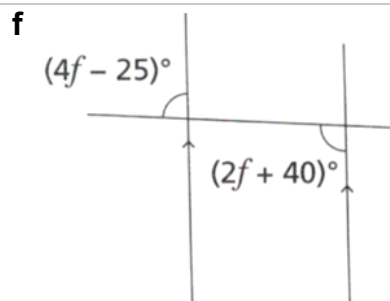
$$c = 31^\circ$$



$$d = 30^\circ$$

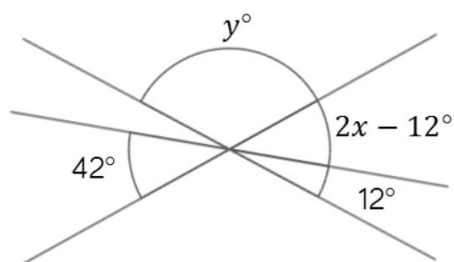


$$e = 37$$



$$f = 27.5^\circ$$

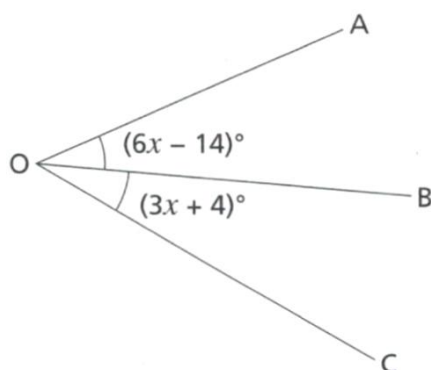
- 7 Calculate the values of x and y .



$$x = 32$$

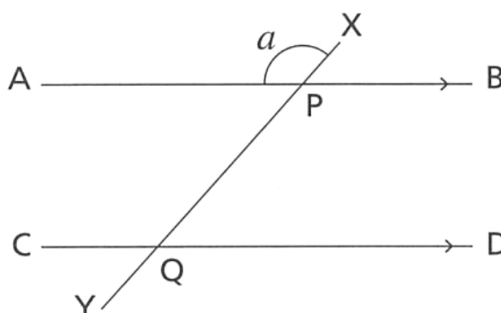
$$y = 116^\circ$$

- 8 OB is the bisector of $\angle AOC$. (in other words, OB splits $\angle AOC$ into equal halves).
Work out the size of $\angle AOC$



$$\angle AOC = 44^\circ$$

- 9 Consider the diagram below.



- a Prove that $\angle CQY = 180^\circ - a$

$$\angle CPX = a \text{ (corresponding angles on } \parallel \text{ lines)}$$

$$\angle CQY = 180^\circ - a \text{ (angles on a straight line)}$$

- b Prove that $\angle DQY = a$

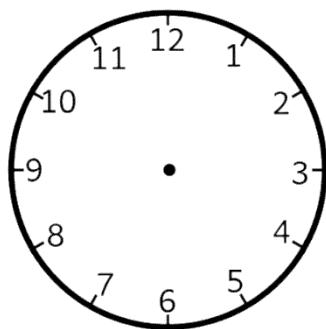
$$\angle CQY + \angle DQY = 180^\circ$$

$$180^\circ - a + \angle DQY = 180^\circ$$

$$\therefore \angle DQY = a$$

Challenge Exercise

- 1 Determine the angle between the minute hand and hour hand at:



a 6:00 pm

180°

b 11:00 pm

30°

c 4:00 am

120°

d 3:30 pm

75°

- 2 How many degrees does the hour hand of a clock turn in:

a 12 hours?

360°

b 24 hours?

720°

c 6 hours?

180°

d 1 hour?

30°

e 30 minutes?

15°

f 1 minute?

0.5°

- 3 Suppose A, B and C are three points on a line, in that order. If $\angle ABX = 40^\circ$, then find $\angle CBX$.

$$\angle ABX + \angle CBX = 180^\circ \text{ (angles on straight line ABC)}$$

$$40^\circ + \angle CBX = 180^\circ$$

$$\angle CBX = 140^\circ$$

- 4 Lines AB and CD intersect at O. If $\angle AOD = 60^\circ$, then find $\angle BOC$.

$$\angle BOC = \angle AOD = 60^\circ$$

(vertically opposite angles)

- 5 Suppose $\angle ABC$ is obtuse. What type of angle is $\angle DEF$, given that $\angle ABC = 2\angle DEF$?

$$\angle ABC \text{ is obtuse, so } 90^\circ < \angle ABC < 180^\circ$$

$$\angle ABC = 2\angle DEF$$

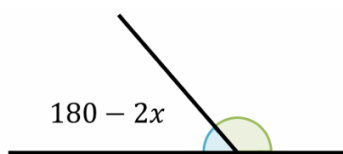
$$\therefore \angle DEF = \frac{\angle ABC}{2}$$

$$45^\circ < \angle DEF < 90^\circ$$

$\angle DEF$ is acute

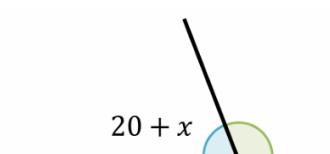
6 Determine the missing angle in each diagram in terms of x or y .

a



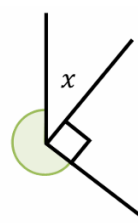
$2x$

b



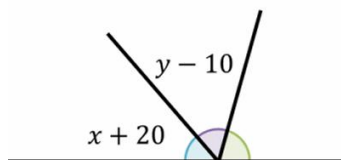
$160 - x$

c



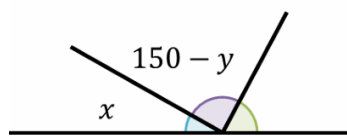
$270 - x$

d



$170 - x - y$

e



$30 - x + y$

7 Suppose $\angle ABC$ is obtuse. What type of angle is $\angle DEF$, given that $\angle DEF = 2\angle ABC$?

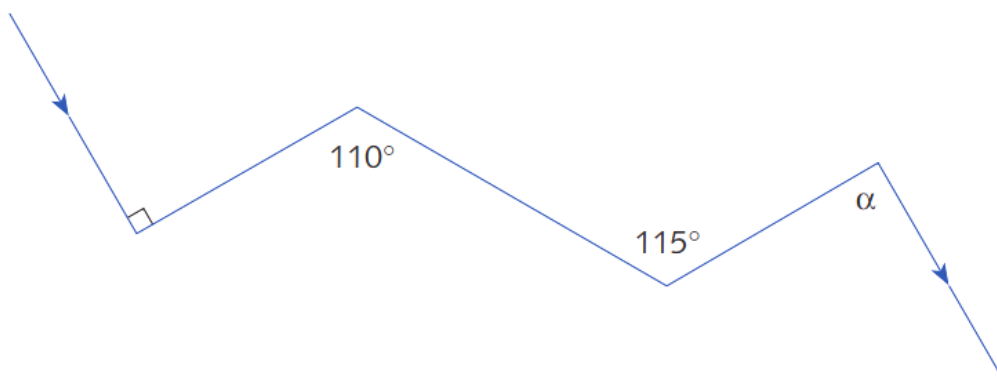
$\angle ABC$ is obtuse, so $90^\circ < \angle ABC < 180^\circ$

$\angle DEF = 2\angle ABC$

$\therefore 180^\circ < \angle DEF < 360^\circ$

$\angle DEF$ is reflex

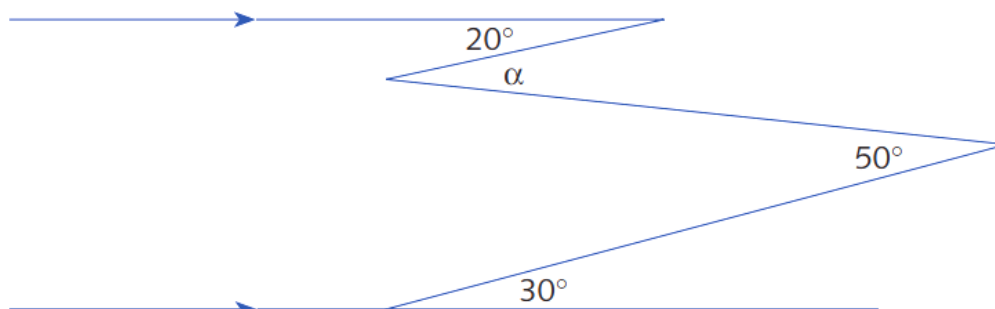
8 Calculate the unknown angle.



95°

9

a Find the value of α .



40°

10 Determine the angle between the hour hand and minute hand of a clock at

a 10:10 am

b 4:45 am

c 7:15 am

d 2:35 p.m

115°

127.5°

127.5°

132.5°

11 Benji thinks that the angle between the hands of a clock at 6:30 pm is 0°. Explain why Benji is wrong and find the correct angle.



The hour hand is not at 6 when it is 6:30.

The two hands are $\frac{1}{24}$ th of a revolution apart, so $\frac{360}{24} = 15^\circ$

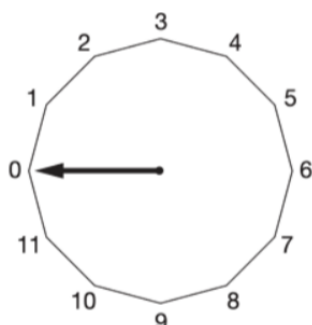
12 You are a mouse. How many mini-cheeses will fit into this container?



Each layer has $360^\circ \div 45^\circ = 8$ cheeses

2 layers of 8, so 16 mini cheeses

13 This regular 12-sided shape has a number at each vertex. Ben turns the pointer from zero, clockwise through 150° . Which number will the pointer now be at?

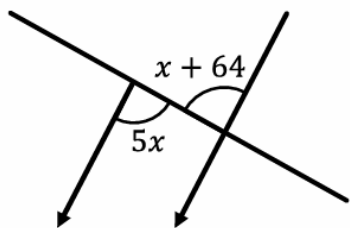


Each section is $360^\circ \div 12 = 30^\circ$

$150^\circ \div 30^\circ = 5$, so number 5.

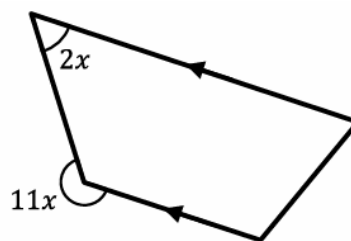
14 Find the value of x .

a



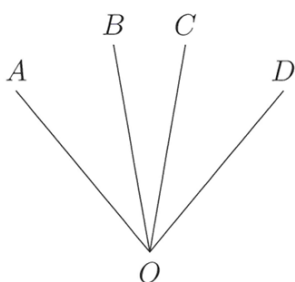
$$\begin{aligned} 5x &= x + 64 \\ 4x &= 64 \\ x &= 16 \end{aligned}$$

b



$$\begin{aligned} 2x + (360 - 11x) &= 180^\circ \\ -9x &= -180^\circ \\ x &= 20^\circ \end{aligned}$$

15 In the figure below, $\angle AOC = 50^\circ$, $\angle BOD = 50^\circ$ and $\angle AOD = 80^\circ$. Find $\angle BOC$.



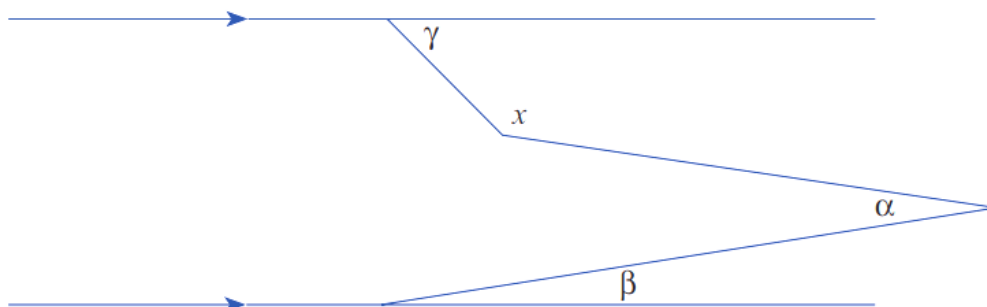
$$\angle AOC + \angle BOC = 100^\circ$$

$$\text{But } \angle AOD = 80^\circ$$

So the overlap must be 20°

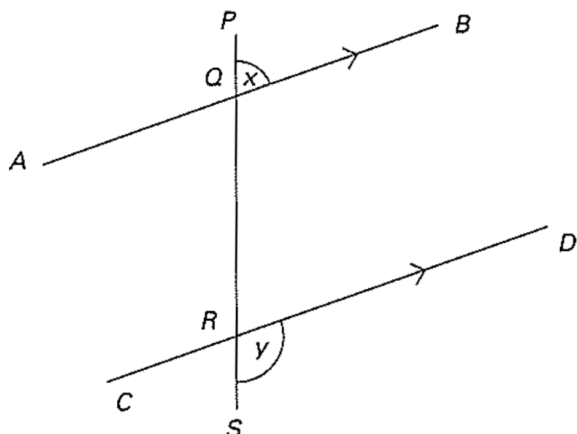
$$x = \gamma + \beta - \alpha$$

16 Find x in terms of α , β and γ .



$$x = 180^\circ - \gamma + \alpha - \beta$$

- 17 AB is parallel to CD and $PQRS$ is a transversal.
If $y = 3x$, find the size of all angles, giving reasons.



$$\angle PQB = \angle QRD = x \text{ (alternate angles since } AB \parallel CD \text{)}$$

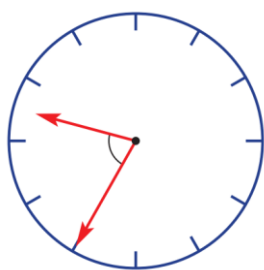
$$\angle QRD + \angle DRS = 180 \text{ (angle sum on straight line QRS)}$$

$$x + 3x = 180$$

$$x = 45 \text{ and } 3x = 135$$

So all angles are either 45° or 135°

- 18 The angle between the two hands of a clock at 9:35 a.m. is 77.5°
Explain why. (hint: how much does the hour hand move in 5 minutes?)



$$\text{Minute hand is at position } (180 + 30) = 210^\circ$$

$$\text{Hour hand is at position } (270 + 35 \times 0.5) = 287.5^\circ$$

$$\text{The difference between the two hands is } 77.5^\circ$$

- 19 What is the obtuse angle between the hands of a clock at 6 minutes past 8 o'clock?

$$\text{At 8, the angle between the hands is } 120^\circ$$

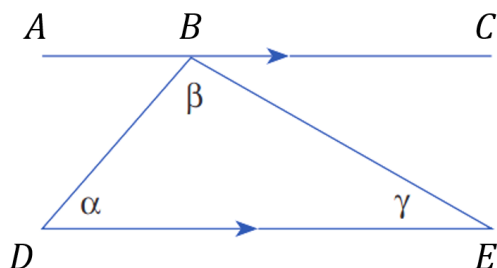
$$\text{In the following 6 minutes, the minute hand turns } 36^\circ \text{ while the hour hand turns through an angle of } 3^\circ \text{ in the same direction.}$$

$$\text{So the obtuse angle between the hands increases by } 33^\circ$$

$$\text{So all angles are either } 45^\circ \text{ or } 135^\circ$$

20

- a Prove that $\alpha + \beta + \gamma = 180^\circ$, that is, prove that the sum of the angles in any triangle is 180°



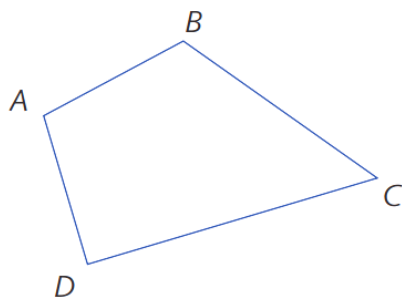
$$\angle ABD = \alpha \text{ (alternate angles on } \parallel \text{ lines)}$$

$$\angle CBE = \gamma \text{ (alternate angles on } \parallel \text{ lines)}$$

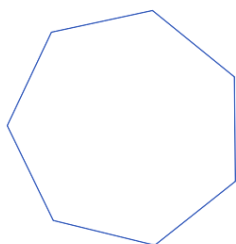
Consider line $\angle ABC$:

$$\alpha + \beta + \gamma = 180^\circ \text{ (angles on a straight line sum to } 180^\circ \text{)}$$

- b** By dividing the quadrilateral below into two triangles, show that the angle sum in a quadrilateral is 360°



- c** A heptagon is a seven-sided figure, as shown below. What is the angle sum of a heptagon?



Split the heptagon into 5 triangles. $180 \times 5 = 900^\circ$

- 21** How many minutes pass after 12 noon before the angle between the minute and hour hand is exactly 90° ?

At time t minutes after 12 noon:

The minute hand moves 6° per minute, so its angle is $6t^\circ$

The hour hand moves 0.5° per minute, so its angle is $0.5t^\circ$

The angle between them is $6t - 0.5t = 5.5t^\circ$

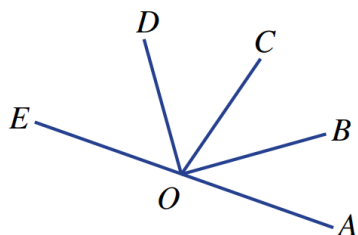
For the first 90° angle occurrence after noon:

$$5.5t = 90$$

$$t = \frac{90}{5.5} = \frac{180}{11} = 16\frac{4}{11}$$

$16\frac{4}{11}$ minutes (or 16 minutes 22 seconds)

- 22** In this diagram $\angle AOB = \angle BOC$ and $\angle COD = \angle DOE$. Give reasons why $\angle BOD = 90^\circ$



$$\angle AOB + \angle BOC + \angle COD + \angle DOE = 180^\circ \text{ (angles on a straight line)}$$

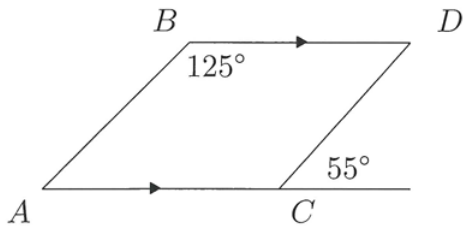
Since $\angle AOB = \angle BOC$ and $\angle COD = \angle DOE$, we can write

$$2\angle BOC + 2\angle COD = 180^\circ$$

$$\angle AOB + \angle COD = 90^\circ$$

$$\therefore \angle BOD = 90^\circ$$

23 Prove that $AB \parallel CD$.

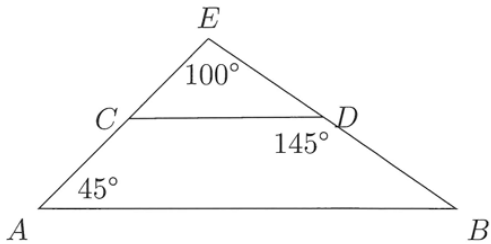


$$\angle BDC = 55^\circ \text{ (alt. angles, } BD \parallel AC \text{)}$$

$$\text{Therefore, } \angle B + \angle C = 125^\circ + 55^\circ = 180^\circ.$$

Therefore, the co-interior angles are supplementary. So $AB \parallel CD$.

24 Prove that $AB \parallel CD$.

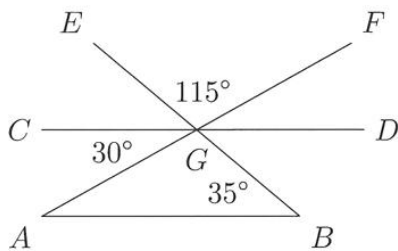


$$\angle BCA = 180^\circ - 45^\circ - 100^\circ = 35^\circ.$$

$$\text{Therefore, } \angle DEC + \angle ACE = 145^\circ + 35^\circ = 180^\circ.$$

Therefore, the corresponding angles are equal. So $AB \parallel CD$.

25 Prove that $AB \parallel CD$.

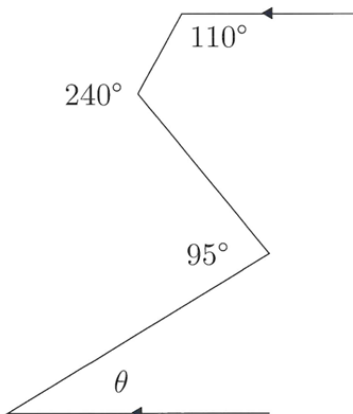


$$\angle EGC = 180^\circ - 30^\circ - 115^\circ = 35^\circ.$$

$$\text{Now note that } \angle EGC = \angle GBA = 35^\circ.$$

Therefore, the corresponding angles are equal. So $AB \parallel CD$.

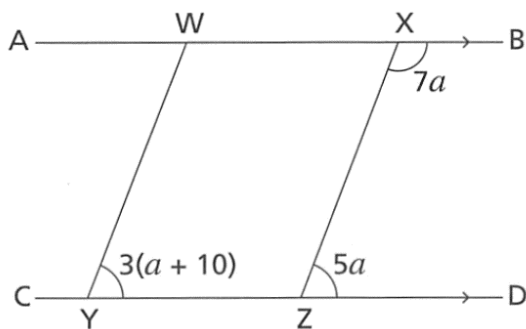
26 Find the value of θ .



Sketch two horizontal parallel lines.

$$\theta = 45^\circ$$

- 27** AB is parallel to CD. Prove that WY is parallel to XZ.



If parallel, then both these equations must be true:

$$3(a + 10) = 5a \text{ (corresponding angles on } \parallel \text{ lines) and}$$

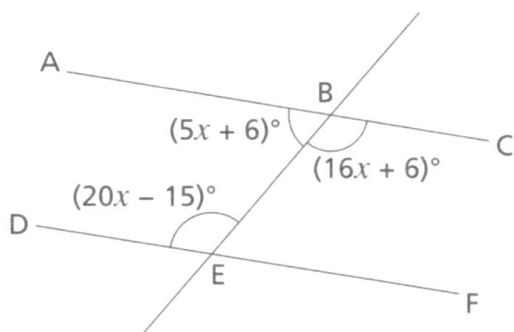
$$5a + 7a = 180 \text{ (co-interior angles in } \parallel \text{ lines)}$$

After we solve for the value of a in the first equation, if this value also works for the second equation, then the lines must be parallel

By solving $3(a + 10) = 5a$ we get $a = 15$

$a = 15$ works for $5a + 7a = 180$, so both lines are parallel

- 28** ABC is a straight line. DEF is a straight line. Is ABC parallel to DEF?
Show working to explain how you know.



If parallel, then both these equations must be true:

$$20x - 15 = 16x + 6 \text{ (alternate angles on } \parallel \text{ lines) and}$$

$$20x - 15 + 5x + 6 = 180 \text{ (co-interior angles in } \parallel \text{ lines)}$$

By solving $20x - 15 = 16x + 6$, we get $a = \frac{21}{4}$

$a = \frac{21}{4}$ does not satisfy $20x - 15 + 5x + 6 = 180$, so not parallel.